THE GLOBAL FUSION INDUSTRY

Fusion Companies Survey by the Fusion Industry Association

FUSION
INDUSTRY ASSOCIATION

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FOREWORD

The transition of fusion energy from the national lab to the marketplace is now firmly underway. With $4.5 billion of new funding in the last 12 months, private companies are now the driving factor in the development of fusion energy. For decades, governments had defined the way to fusion, through national labs and government R&D programs. Now, they are racing to write and revise their fusion strategies to take advantage of the transition, moving timelines forward, devising public-private partnerships that allow knowledge sharing, and adapting their infrastructure to not only support the growth of the new industry in their region, but to lead globally.

Andrew Holland - CEO
Fusion Industry Association

When we first published this report in 2021, this trajectory was not the obvious outcome. With less than $2 billion invested in the sector, and 85% of that in just four companies, the private sector was emerging as an interesting new player, but governments still dominated. Since then, we have seen a 650% increase in industry investment. The $4.5 billion invested in the industry this year alone dwarfs any single government fusion budget for the same timeframe. The result is a growing fusion workforce comprising over 16,000 employees, rapid construction of new facilities, and a maturing supply chain able to respond to fusion's growing demands.

While companies focus on achieving their technical and commercial milestones, governments should address common challenges across the industry - in particular, commercially-relevant science in resilient materials and the fusion fuel cycle. In the last two years, leading governments have updated their fusion strategies and roadmaps to reflect the new reality. In this increasingly competitive marketplace, governments that maintain their programs and funding priorities along 2021 lines will fail to capitalize and attract the emerging industry.

What hasn't changed is the timeline. Since the first report, the majority of fusion companies have continued to report that their goal is to produce commercial energy from fusion in the 2030s, with most saying that they expect to see that goal hit in the early 2030s. These timelines remain consistent, which is notable for those who've been in fusion for a while! With sufficient resources, the industry is confident that fusion will meet its goals.

New hunger for energy is driving fusion

In the end, the prize is clear. A commercial fusion industry is a world-changing industry, and the returns on investment will be massive. It will be the industrial base of the 21st century. In earlier years, when introducing this report, I identified environmental threats and energy security demands as important drivers of fusion development. These remain, but the last year has added an important new driver: the energy demand of artificial intelligence. The need for a source of clean energy that can provide large amounts of power to drive the data centers fueling the AI revolution has only grown. Firms that are building the massive datasets required for more powerful AI models (the hyperscalers) are faced with the problem of how to power these energy-hungry chips that create the models. This is a business problem as well as a political problem, as energy consumers around the world are feeling the cost of rising energy prices every day.

Fusion will be the solution. Its approach of turning abundant, energy dense fuel into power, without carbon emissions, make it the only source of energy that will be able to scale up to meet constantly growing demand. Even though it cannot deliver today, the demand from the market is bringing fusion forward. We see this in the data in this report. For the first time this year, we asked whether companies had siting and power purchase agreements for a commercial power plant. Six already have a siting agreement and four more are actively evaluating options. Five had a power purchase agreement, offtake agreement, or similar commercial offtake commitment, while two more were in discussions. These agreements provide the baseline for financing a fusion power plant, reducing market risk by ensuring that there will be buyers as soon as fusion has power to sell.

The importance of setting milestones

As companies and countries race to fusion, it is important for observers to be able to track where the players stand in the race. This reflects an important principle: to be able to raise new funding, companies must be open and clear, at least with their investors, about what they are aiming for, what their milestones are, and whether they are achieving them.

Because the pathways to commercial fusion are diverse, these milestones will inevitably differ from company to company. Some firms are pursuing revenue-generating intermediate applications, while others are spinning out subsidiary businesses or focusing exclusively on licensing their technology.

While peer review is the standard in the scientific community, that doesn't mean that the commercial community doesn't also have its version of peer review: the global marketplace, which in some ways, can have a far harsher discipline. The FIA has recently published a new page on our website outlining the milestones already hit in fusion, and the pathway fusion companies are following to achieve deployment: www.fusionindustryassociation.org/fusion-is-coming

Pioneering entrants into public markets

As we publish this report, two private fusion companies are in the process of joining the public markets through mergers. Participating in the stock market will provide access to new investors, but will also bring a different type of scrutiny to fusion, with the day-to-day financial progress of companies able to be tracked simply on a minute-to-minute ticker symbol. It expands access to capital, allows a broader range of investors to participate in fusion's growth, and demonstrates increasing confidence that fusion is moving from a scientific endeavor toward a commercial energy industry.

The role public markets can play in scaling transformative technology can be vital. The growth of the semiconductor, biotechnology, internet, and renewable energy sectors was accelerated by access to public capital that enabled companies to finance large facilities, build supply chains, hire talent, and bring products to market at scale.

Fusion companies that can tap those markets can help with those key challenges, as they provide an important mechanism for channeling long-term investment. Recent increases in value to companies in other energy and technology sectors make the allure of access to the public markets an obvious choice for some. The arrival of fusion companies on public exchanges is a sign that the industry is entering a new era. It is now up to all of us in the industry to deliver.

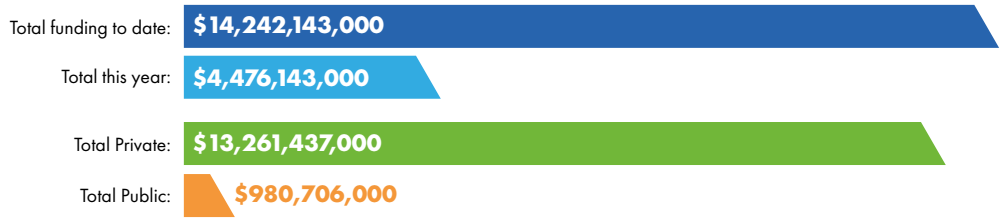
Conclusions: Fusion is on its way to commercialization

Two years ago, in the introduction to our 2024 report, I wrote that "Ambitious timelines require ambitious resources," as I explained after a difficult year in funding that fusion would not meet its goals of commercial fusion in the 2030s with increases of only $1 billion new funding per year. Fortunately, that year was a product of broader difficult market conditions, and not an indicator of fusion's pathway. Today, with over $4 billion invested in the last year, I am confident that the industry is now being given the private investment it needs to get across the line. With that increased investment comes increased scrutiny. Fusion companies are setting bold goals. As we enter the second half of the decade, companies will start to deliver on those goals. For companies that cannot meet their objectives, market consequences will follow. When we emerge into a fully commercial fusion industry, few expect there to be 56 companies in it. The race is on.

2026: THE HEADLINE FIGURES

©Tokamak Energy

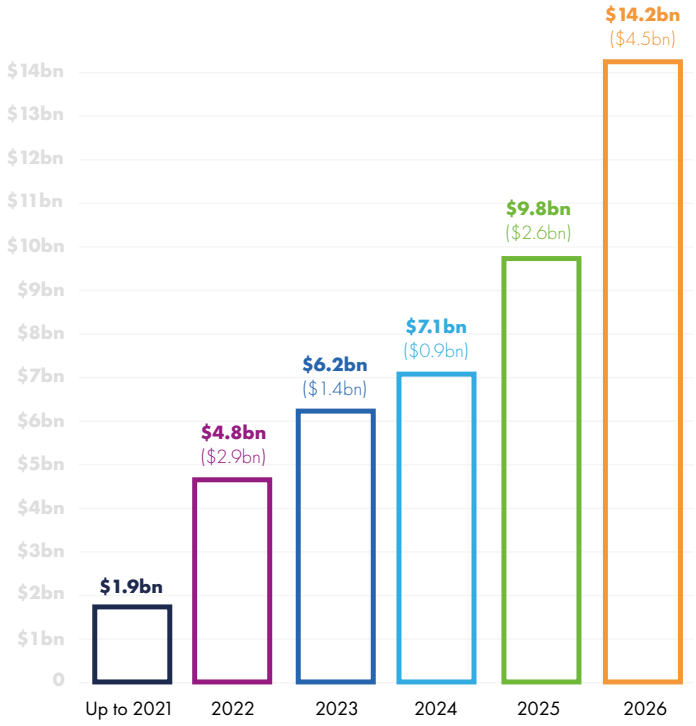
Total industry funding in 2026



*Some reported figures have been converted to dollars/rounded.

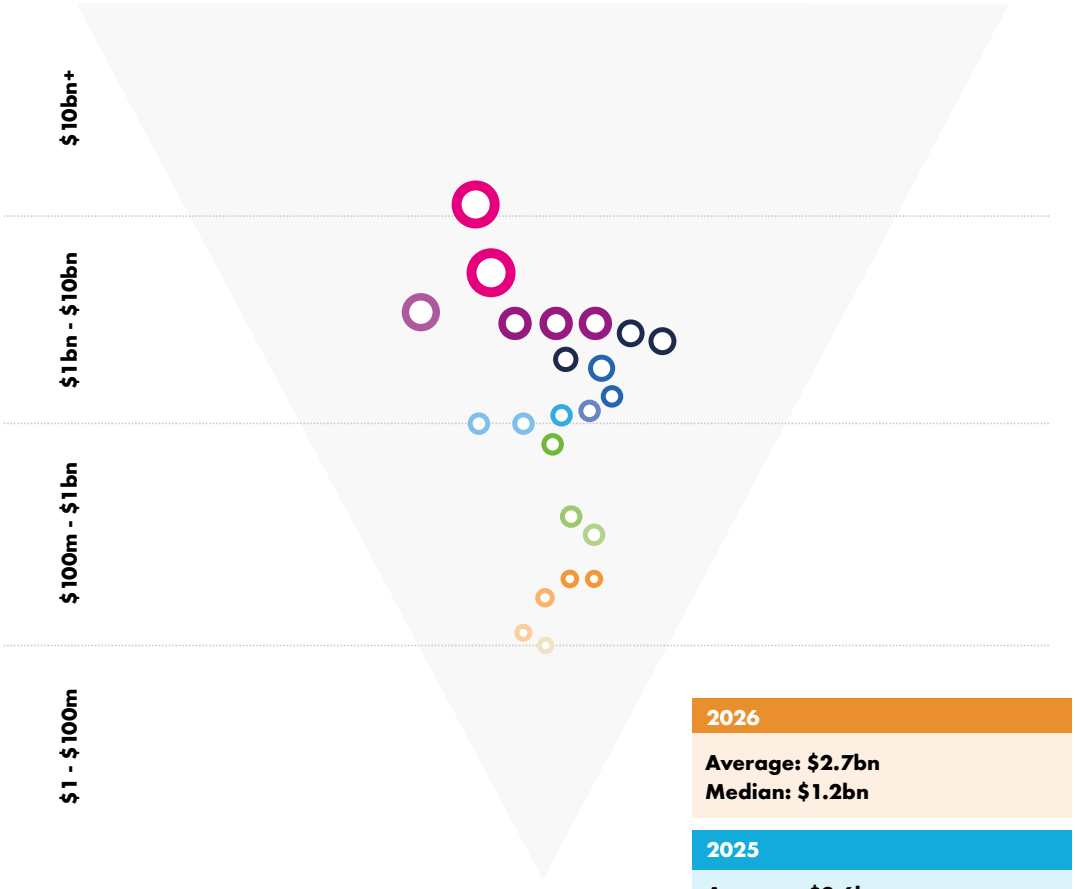
Total industry funding over the years

(annual total in brackets)



How much more funding do you estimate needing to deliver a commercial plant?

23 responses



2026

Average: $2.7bn
Median: $1.2bn

2025

Average: $2.6bn
Median: $694m

THE FUSION INDUSTRY IN 2026

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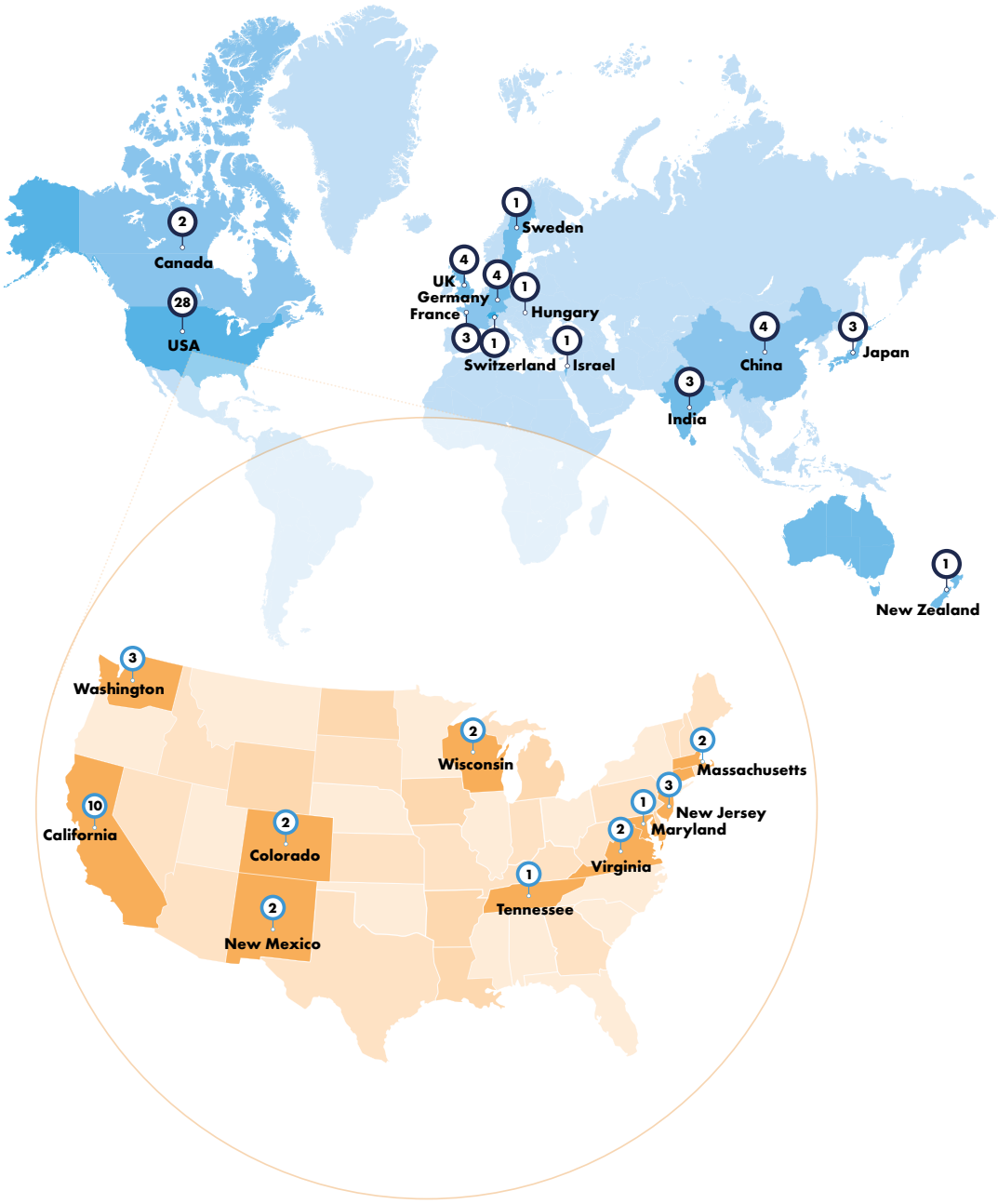
1. Location

By primary HQ



4. Selected* investors in fusion in the last 12 months

Al Mirqab Capital
Ankur Capital
Aoki Super
Bessemer Venture Partners
Bhukhanwala Industries
Blast Club
BMH Hessen
BPI France
Brevan Howard Macro Venture Fund
Congruent
Counterpoint Global (Morgan Stanley)
CREST SKILL Partners
Crédit Mutuel Impact
Daiwa House Ventures
Daphni
DEEPCORE
Deerfield
Development Bank of Japan
East X Ventures
Ecrowd NEXT
ENN Group
European Innovation Council Fund
German Federal Agency for Disruptive Innovation (SPRIND)
Fidelity
Founders Fund
FFA Private Bank
Fujikura
Galaxy Interactive
Gigascale Capital
Google Ventures
HAKOBUNE
Hasetora
HOF Capital
Holystar Electrical Tech
Host Plus
Icehouse Ventures
Industrial47
IP Group
IQT
JERA Co.
JGC JAPAN CORP.
KDDI Green Partners Fund
Keio Innovation Initiative
Kinokisen
Konoike Group
Kyocera Venture Innovation Fund-I
Laukien Visionary Investment Fund
Leave a Nest Capital
Leitmotif
Lightspeed
Long Journey
Lowercarbon Capital
Marseille Innovation
Mercia Capital Management
Mitani Group
Mitsubishi Corp
Mitsui & Co
Mitsui Fudosan
Mitsui O.S.K. Lines
Mizuho Capital
Modern Capital
Neo
Neva SGR (Intesa Sanpaolo Bank)
Nichias
Nissay Capital
Noshiro Densetsu Kogyo
NTT
NVentures
Oaktree
OKANO VALVE MFG
OKB Capital
Outset Ventures
Overlay Capital
Pelican Energy Partners
pi Ventures
Planet First Partners
Playfair
Prime Impact Fund
Prime Movers Lab
RA Capital
RWE
SBI Investment
Silicon Roundabout Ventures
Speedinvest
STATION Ai Corp
Stanley Druckenmiller
Stonehill Technology
Sumitomo Mitsui
Tees River
The Kansai Electric Power Company
Threshold Ventures
Tokai Tokyo Investment
Toyota Ventures
Toyoda Gosei
Uncork Capital
WndrCo
Woori Venture Partners
X&KSK I Investment Limited Partnership
Yamada Shokai Holding
YNF Limited

* All of these investors have been publicly identified. The FIA is not responsible for the responses listed in this report from survey participants and does not intend to disclose any proprietary information.

7. Approach

56 responses

Magnetic confinement

- Stellarator: **8**
- Tokamak: **5**
- Spherical tokamak: **3**
- Field Reversed Configuration: **3**
- Gas Dynamic Magnetic Trap: **1**
- Centrifugal Magnetic Mirror: **1**
- Open Magnetic confinement: **1**
- Pulsed Stellarator: **1**
- Tandem magnetic mirror: **1**
- Levitated Dipole: **1**
- Levitron: **1**

Inertial confinement

- Laser-driven inertial confinement: **9**
- Quantum enhanced laser-driven inertial confinement: **1**
- Hybrid Fast Ignition: **1**
- Pulser-driven inertial confinement: **1**

Magneto-inertial

- MagLif: **1**
- Magnetized target fusion: **1**
- Magnetized target impact fusion: **1**
- Plasma jet driven magneto-inertial fusion: **1**
- Laser-driven magneto-inertial fusion: **1**
- Plectoneme: **1**
- Dense Plasma Focus: **1**
- Z-pinch: **1**
- Sheared-flow-stabilized Z-pinch: **1**

Hybrid electrostatic

- Magnetic-electrostatic confinement: **3**
- Electrostatic with a plasma cusp: **2**
- Inertial-electrostatic lattice confinement: **1**
- Augmented Beam-target fusion: **1**

Muon catalyzed fusion

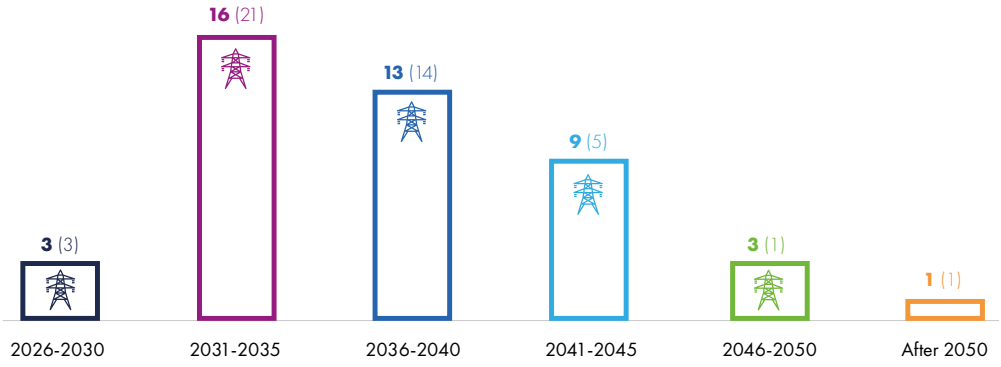
- Muon catalyzed high density fusion cell: **1**

10. Predictions

When will the first fusion plant deliver commercial electricity?

(Last year's responses in brackets)

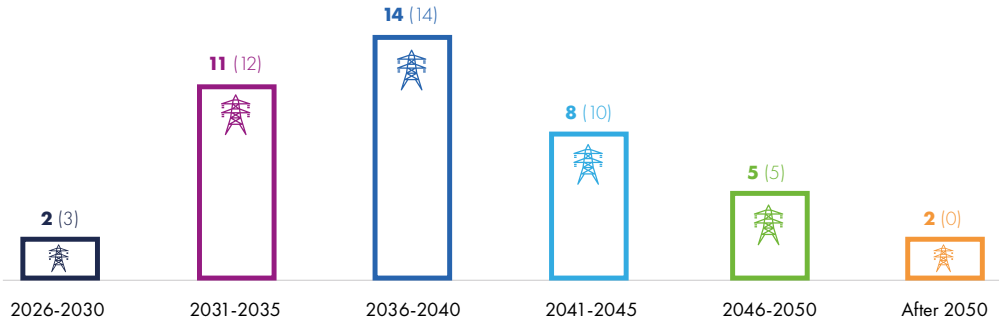
45 responses



When will the first fusion plant demonstrate a low enough cost/high enough efficiency (Q) to be considered commercially viable?

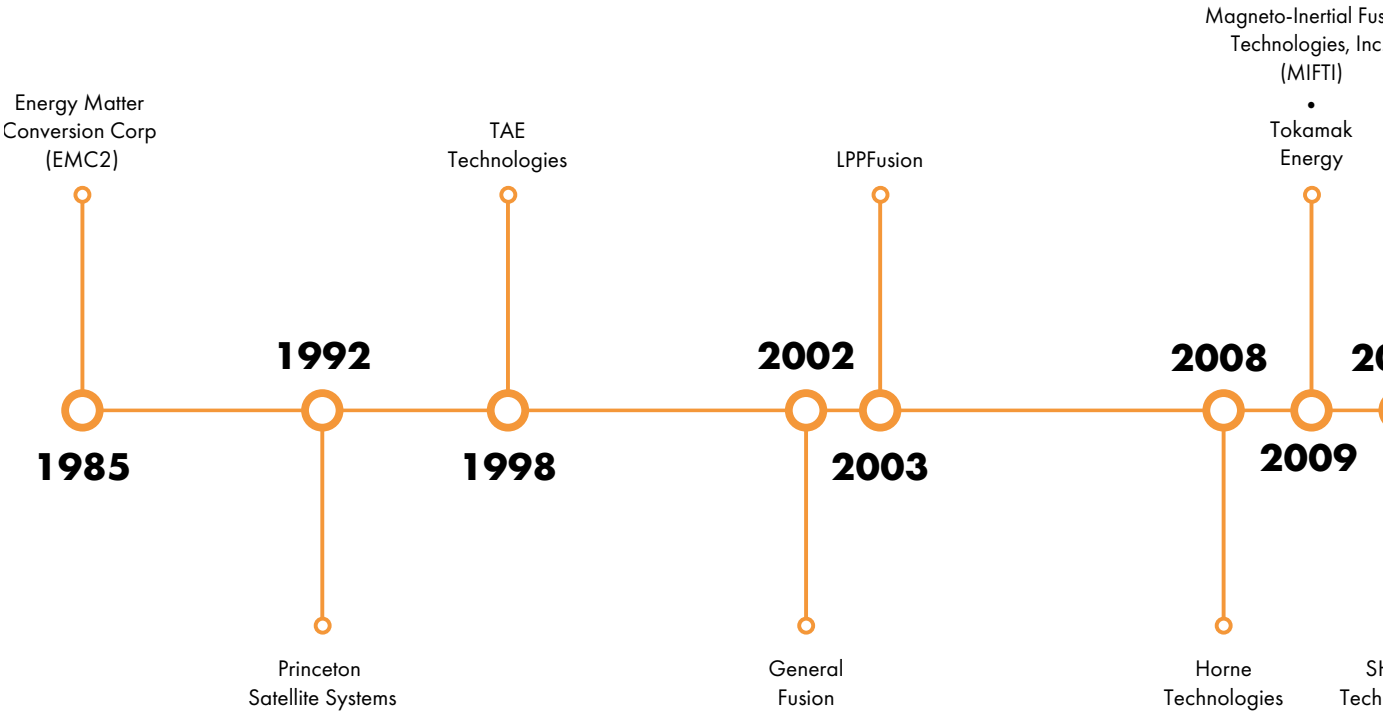
(Last year's responses in brackets)

42 responses

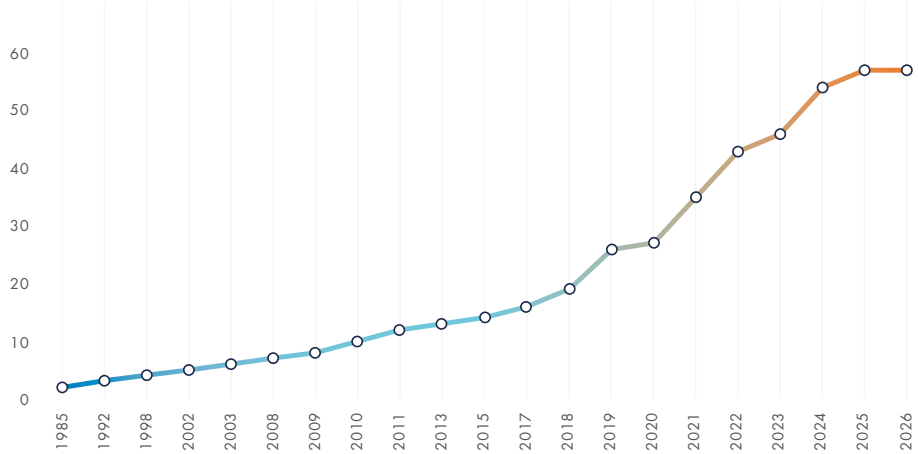


12. Timeline of companies founded

Companies new to the survey in orange



Total number of private fusion companies by year



PROFILES OF TODAY'S FUSION PLAYERS

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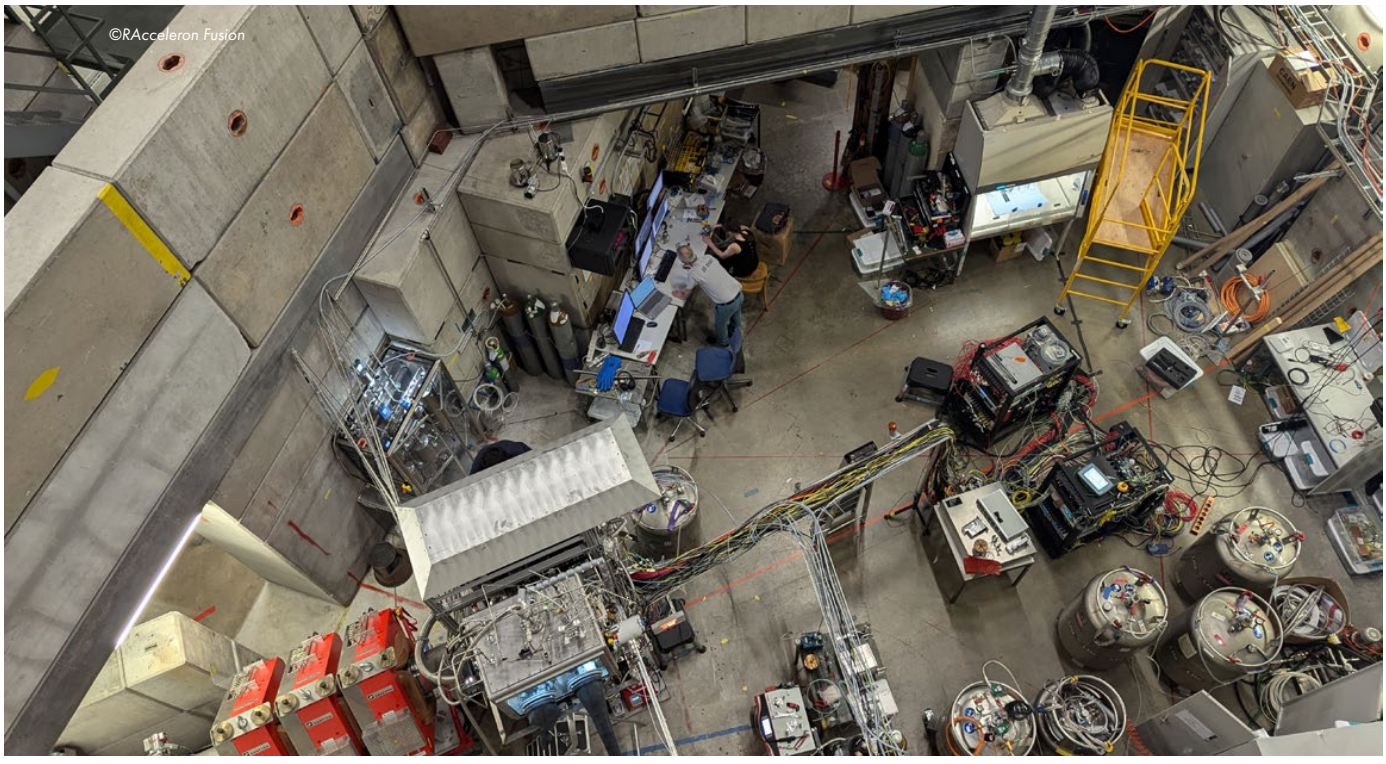
acceleron

FIA MEMBER

ACCELERON FUSION

Acceleron Fusion is developing muon-catalyzed fusion as an abundant new source of clean energy.

Primary location	Cambridge, Massachusetts, USA
Contact details	info@acceleron.energy
Year founded	2023
Founder names	Ara Knaian, Seth Newburg
Primary target market(s)	Electricity generation
Total declared funding	$26,500,000
Employees	20
General approach	Muon-catalyzed fusion
Specific approach	Active-target muon production and high density fusion cell
Fuel Source	DT
Planned energy capture approach	Lithium neutron 'blanket'
Target: First pilot plant	Early 2030's
Anticipated MWe of your commercial operating facility	200 MWe


©RAcceleron Fusion

ASPL FUSION

Agnira Sanlayan Private Limited (ASPL) Fusion is India's leading private nuclear fusion company, incorporated in August 2025. It is being incubated at Institute for Plasma Research (IPR), Department of Atomic Energy (DAE), India. We pursue a phased magnetic-mirror roadmap towards a sub critical fusion-fission hybrid and pure-fusion electricity, with near-term revenue streams via BNCT and NDT/NAA channels.

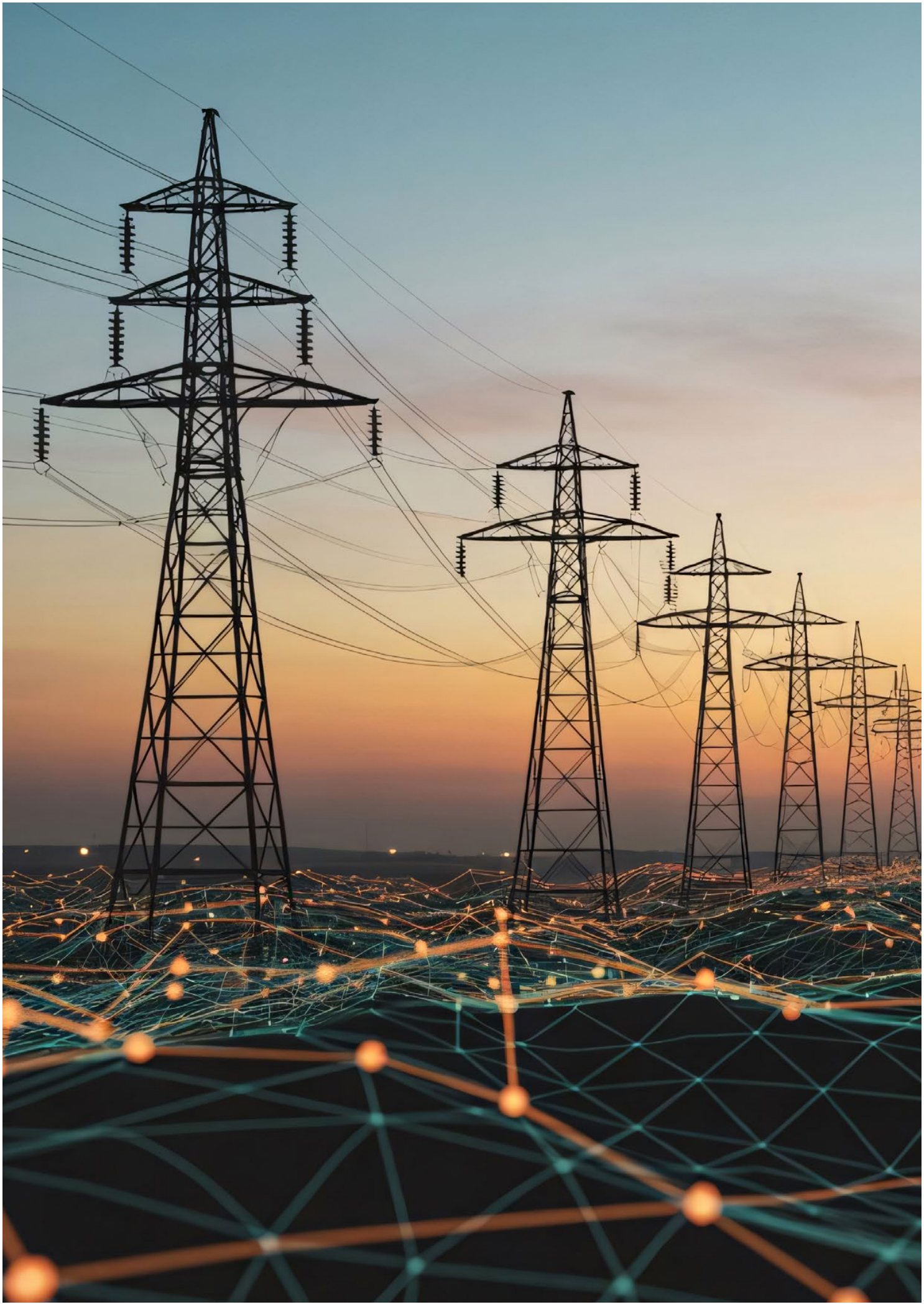
Primary location	Gandhinagar, Gujarat, India
Contact details	ceo@asplfusion.com
Year founded	2025
Founder names	Prof Prabhat Ranjan, Mr Abhay Jha and Mr Abhinav Jha
Primary target market(s)	Electricity generation, Medical
Total declared funding	$2,000,000
Employees	7
General approach	Magnetic Mirror
Specific approach	Gas Dynamic Magnetic Trap (GDMT) / Gas Dynamic Trap (GDT)
Fuel Source	D-T
Planned energy capture approach	Lithium neutron 'blanket'
Target: Net fusion energy gain (scientific)	2031-36
Target: Net energy gain (engineering)	2036-41
Target: First pilot plant	2041
Anticipated MWe of your commercial operating facility	250 MWe
Milestones in past 12 months	1) Company incorporated in August 2025 and incubation process started at the IPR, Gandhinagar. 2) Secured operational site in GIDC, Gandhinagar
Interim plants or facilities planned prior to your commercial pilot plant?	1) PRABHA-BNCT: accelerator-based clinical BNCT facility with additional shared but independent NDE vault. 2) PRABHA- HTS: Volumetric neutron-source based on magnetic mirror technology for isotopes production. 3) PRABHA-HYBRID: Sub-critical fusion-fission hybrid reactor with multiple applications including 35 MW of power production. 4) PRABHA-ENERGY: Pure fusion power plant with a target of 250 MW.
Major investments made in past 12 months	1) Purchased site for initial experimental facilities. 2) Set-up R&D office at Infocity in Gandhinagar.
Key collaborators/partners	Being incubated at IPR, DAE, India. Ongoing discussions with industries and academia.

©Astral Systems

AVALANCHE ENERGY

Avalanche Energy is developing a modular 100 kWe to 5 MWe range of compact fusion machines, based on the Orbitron plasma confinement scheme for hard-to-decarbonize applications. Its compact size and modularity are game changers for dual-use, mobile and distributed power applications across air, land, sea, and space. Some potential applications include islanded micro-grids in austere/remote environments, electric vehicle battery recharging, and spacecraft power and propulsion.

Primary location	Seattle, WA
Contact details	reachout@avalanche.energy
Year founded	2018
Founder names	Robin Langtry & Brian Riordan
Primary target market(s)	Electricity generation, Space propulsion, Marine propulsion, Off-grid energy
Total declared funding	$104,200,000
Employees	55
General approach	Magnetic Mirror
Specific approach	Magnetic-electrostatic confinement
Fuel Source	DT
Planned energy capture approach	Lithium neutron 'blanket'
Target: Net fusion energy gain (scientific)	2029
Target: Net energy gain (engineering)	2029
Target: First pilot plant	2030
Anticipated MWe of your commercial operating facility	0.1 MWe
Milestones in past 12 months	1) Stable plasma formation 2) Pure hydrogen plasma 3) 1 keV Plasma Temperature
Interim plants or facilities planned prior to your commercial pilot plant?	FusionWERX commercial test facility for developing and qualifying fusion technologies and hardware.
Major investments made in past 12 months	FusionWERX test facility including tritium handling equipment and superconducting magnets for future fusion neutron test stands.
Key collaborators/partners	DARPA, DIU, AFWERX, WA State, Fusion Fuel Cycles (FFC)
Spin outs/patents/commercial innovations	High Voltage Feedthroughs, Alphavoltaics, High Entropy Alloys
Published papers from last 12 months	1) Hammerhead: a compact 300 kV vacuum bushing, Nature 2) Mode-enhanced ion loading in a 100 kV orbitrap, AIP Physics of Plasmas 3) An Overview of Vacuum Insulation Technology for the Orbitron Fusion Device, IEEE 1st International Symposium on Discharges and Electrical Insulation in Vacuum



Commonwealth Fusion Systems

FIA MEMBER

COMMONWEALTH FUSION SYSTEMS

Commonwealth Fusion Systems is the world's leading and largest private fusion company. The company's marquee fusion project, SPARC, will generate net energy, paving the way for limitless carbon-free energy. The company has raised close to $3 billion in capital since it was founded in 2018.

Primary location	Devens, Massachusetts, USA
Contact details	info@cfs.energy
Year founded	2018
Founder names	Bob Mumgaard, Dan Brunner, Brandon Sorbom, Dennis Whyte, Martin Greenwald, and Zach Hartwig
Primary target market(s)	Electricity generation
Total declared funding	$3,000,000,000
Employees	1,000
General approach	Magnetic confinement (Tokamak, Stellarator)
Specific approach	Tokamak
Fuel Source	DT
Planned energy capture approach	Lithium neutron 'blanket'
Target: Net fusion energy gain (scientific)	2027
Target: First pilot plant	Break ground on the Fall Line Fusion Power Station in 2027, first power on the grid in early 2030s
Anticipated MWe of your commercial operating facility	~400MWe
Siting agreement for a commercial plant	Joint development agreement with Dominion Energy and a conditional use permit for Fall Line Fusion Power Station in Chesterfield County, VA
Power purchase agreement, offtake agreement, or similar commercial offtake commitment	PPA's with Google and Eni
Milestones in past 12 months	1) SPARC vacuum vessels onsite and under assembly, SPARC utilities wing undergoing commissioning 2) First TF magnet produced at CFS factory in SPARC Tokamak Hall for assembly 3) Fall Line Fusion Power Station received Conditional Use Permit (zoning) and applied for PJM interconnection
Interim plants or facilities planned prior to your commercial pilot plant?	1) SPARC is the world's first commercially relevant fusion energy machine to produce more energy from fusion than it needs to power the process. SPARC will derisk 13 of 17 systems for the ARC power plant. 2) FLiBe (Lithium Fluoride-Beryllium Fluoride) R&D thermal loop for the ARC power plant, utilizing molten salt for heat transfer and power generation.
Major investments made in past 12 months	Assembly of the SPARC machine is ongoing, with assembly ~80% complete and commissioning of subsystems underway.

Crossfield Fusion

CROSSFIELD FUSION LTD.

Primary location	London, UK
Contact details	enquiries@crossfieldfusion.com
Year founded	2019
Founder names	James McKenzie, Chris Macdonald-Bradley
Primary target market(s)	Fusion-industry relevant neutron source, including isotope manufacture, fusion-energy spectrum relevant materials testing, tritium breeding and breeder blanket testing
Total declared funding	$500,000
General approach	Augmented beam-target systems
Specific approach	Beam-target methodologies based on results of early stage research reactors
Fuel Source	DD, with DT from prompt breeding
Planned energy capture approach	IP in progress (with options analysed for lithium, liquid metals, fluoride salts and/or fission/fertile breeding)
Anticipated MWe of your commercial operating facility	Initial objective; 10^15 neutrons/s/cm^2
Milestones in past 12 months	Breeder blanket test stand prototype design; commenced
Interim plants or facilities planned prior to your commercial pilot plant?	The interim objective is for a peta.neutron/s class source for isotope/tritium breeding and medical applications, as a first realistic step towards stand-alone energy reactors.

ENERGY MATTER CONVERSION CORP (EMC2)

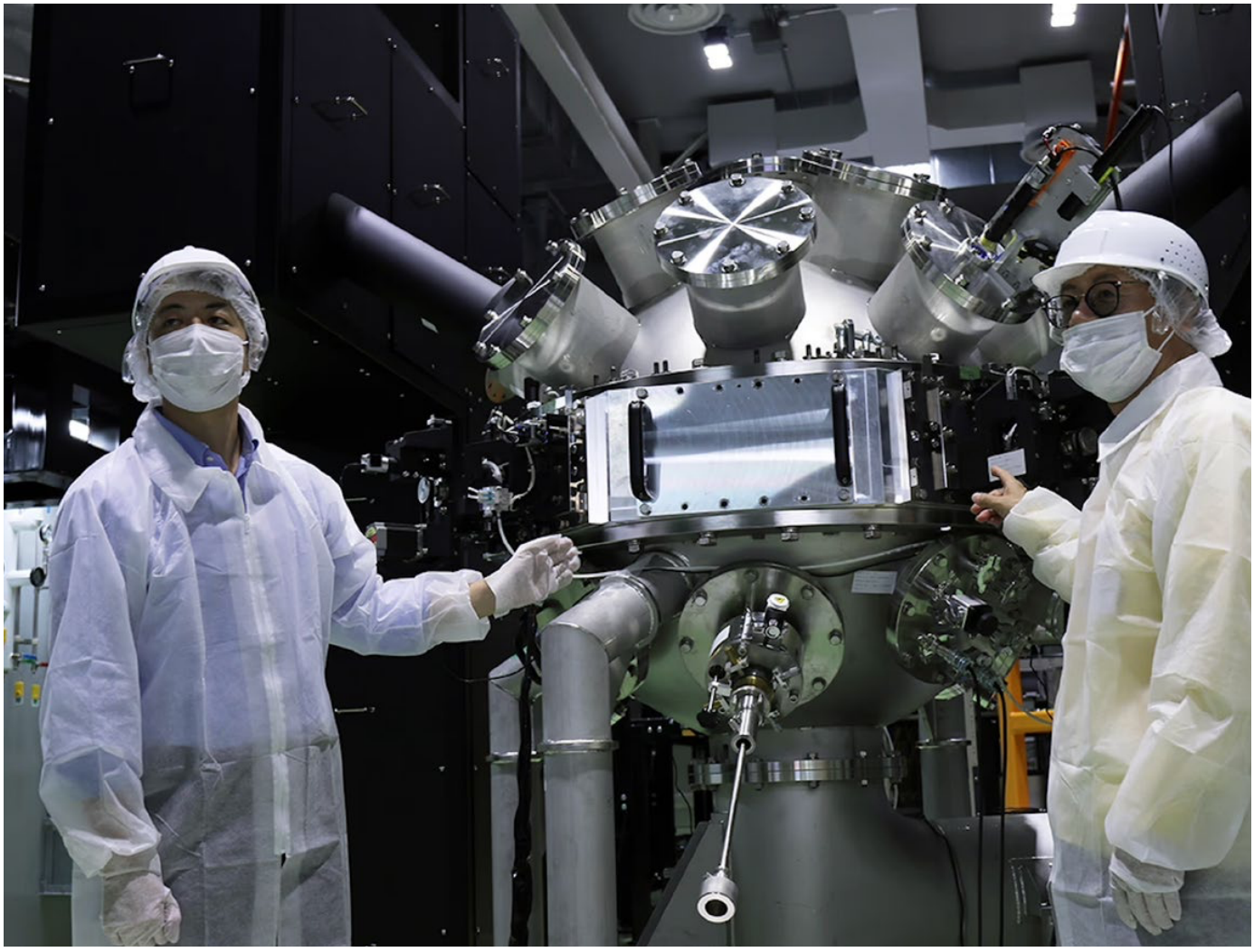
EMC2 pioneers Polywell Fusion for electricity, neutron sources, and propulsion, backed by 40 years of expertise. Its stable magnetic cusp enables high-density, compact operation with natural divertors. This provides low-cost, scalable solutions for fusion energy and high-flux neutron production.

Primary location	San Diego, USA
Contact details	info@emc2fusion.com
Year founded	1985
Founder names	Dr Robert Bussard
Primary target market(s)	Electricity generation, 14MeV Neutron Source, Materials Testing, Nuclear Waste Transmutation
Total declared funding	$40,000,000
Employees	5
General approach	Magneto-Electrostatic
Specific approach	High-beta cusp plasma combined with an electrostatic potential well
Fuel Source	DT
Planned energy capture approach	Lithium neutron 'blanket'
Target: Net fusion energy gain (scientific)	2033
Target: Net energy gain (engineering)	2035
Target: First pilot plant	2038
Anticipated MWe of your commercial operating facility	100-500 MWe
Milestones in past 12 months	1) Updated on Polywell fusion scaling using 6D PIC simulations in a peer reviewed publication 2) Filed two patent applications in the areas of Polywell power plant operation, high-flux neutron source, and Polywell plasma thrusters 3) Completed the development of 6D PIC simulation capability for future Polywell device design and optimization
Interim plants or facilities planned prior to your commercial pilot plant?	Ion beam driven Polywell DT fusion neutron source facility for FPNS/VNS.
Major investments made in past 12 months	Completed the development of 6D PIC simulation capability for future Polywell device design and optimization
Key collaborators/partners	SHINE Technologies, Korea Atomic Energy Research Institute Princeton Plasma Physics Laboratory, Los Alamos National Laboratory, KU LEUVEN, EHT Ventures
Spin outs/patents/commercial innovations	Filed two patent applications in the areas of Polywell power plant operation, high-flux neutron source, and Polywell plasma thrusters.
Published papers from last 12 months	"Polywell Revisited" - Journal of Fusion Energy (2026)

ENN SCIENCE AND TECHNOLOGY DEVELOPMENT CO., LTD.

with HTS magnets.

Primary location	Langfang, China
Year founded	2017
Founder names	Yusuo WANG
Primary target market(s)	Electricity generation
Total declared funding	$720,000,000
Employees	291
General approach	Magnetic confinement (Tokamak, Stellarator)
Specific approach	Spherical tokamak
Fuel Source	pB11
Planned energy capture approach	Direct energy conversion
Target: Net fusion energy gain (scientific)	2030
Target: Net energy gain (engineering)	2035
Target: First pilot plant	2035
Anticipated MWe of your commercial operating facility	1,000-1,200 MWe
Milestones in past 12 months	Through a series of experiments, EXL-50U has achieved several key milestones: the generation of 1 MA plasma current using hydrogen–boron fuel; the first high-confinement mode in proton–boron plasma; sustained electron temperatures of 100 million degrees; and sustained ion temperatures of 50 million degrees.
Interim plants or facilities planned prior to your commercial pilot plant?	EHL-2 will test and verify scientific and technological solutions for p–B11 fusion plasmas, followed by EHL-3 to test and demonstrate engineering solutions for reactor-scale p–B11 burning plasmas.
Major investments made in past 12 months	EXL-50U experiment and upgrade
Published papers from last 12 months	1) Overview of EXL-50 research progress, Nuclear Fusion 2) Experimental and simulation studies on the 12 T D-shaped high temperature superconducting toroidal field coil for compact fusion reactor applications, Superconductor Science and Techonology 3) Active optical boundary recognition with boron powder injection in a magnetic confinement device, Scientific Reports



FIA MEMBER

FIRST LIGHT FUSION LIMITED

With FLARE, First Light is lowering the barrier to economic fusion energy: a development pathway that is rapidly reducing risk at low cost; a simplified, high Q, tritium-rich power plant that sidesteps many challenges of Inertial Fusion; all de-risked by an adjacent commercial pathway that monetises the fusion enabling technologies.

Primary location	Oxford, England
Contact details	enquiries@firstlightfusion.com
Year founded	2011
Founder names	Professor Yiannis Ventikos and Dr Nicholas Hawker
Primary target market(s)	Electricity generation
Total declared funding	$156,379,800
Employees	75
General approach	Inertial confinement
Specific approach	Hybrid Fast Ignition
Fuel Source	DT
Planned energy capture approach	Liquid metal with heat exchanger
Target: Net fusion energy gain (scientific)	Early 2030s
Target: Net energy gain (engineering)	Early 2030s
Target: First pilot plant	Late 2030s
Anticipated MWe of your commercial operating facility	333 MWe
Milestones in past 12 months	1) Publication of FLARE Hybrid Fast Ignition scheme White Paper 2) Validation of TBR of 1.8 in natural isotope lithium 3) Isentropic compression demonstration on BFG
Interim plants or facilities planned prior to your commercial pilot plant?	Extension of in-house M3 facility to include laser capability for integrated FLARE experiments. Low Power Pulsed Power / Laser hybrid Gain facility (with partners).
Key collaborators/partners	Sandia National Laboratory, Imperial College London, University of Oxford, University of York, STFC.
Spin outs/patents/commercial innovations	FLARE patent filings (core system and power plant). Commercial sales of VIPER velocity amplifier for hyper velocity (space) relevant testing. RODA - Robust Optimisation Design Accelerator - AI tool released to market.
Published papers from last 12 months	1) A Bayesian approach to time-domain Photonic Doppler Velocimetry analysis, Review of Scientific Instruments. 2) Equation of state measurements of additively manufactured tantalum up to 437 GPa, Journal of Applied Physics. 3) A Multi-MA Pulsed Power Facility for Implosion Hydrodynamics Experiments

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FOCUSED
FOCUSED

FUSE

FIA MEMBER

FUSE

Fuse's mission is to ensure US energy dominance through fusion energy while providing radiation-as-a-service now for the U.S. Department of War for both strategic and conventional systems, as well as critical infrastructure and commercial customers. The technology that allows Fuse to provide radiation resilience testing today is the same technology delivering fusion tomorrow.

Primary location	San Leandro, California, USA
Contact details	jc@f.energy
Year founded	2019
Founder names	JC Btaiche
Primary target market(s)	Electricity generation
Total declared funding	$65,000,000
Employees	60
General approach	MagLif
Specific approach	MagLif
Fuel Source	DD, DT
Planned energy capture approach	NA
Anticipated MWe of your commercial operating facility	50 MWe
Interim plants or facilities planned prior to your commercial pilot plant?	Facilities in San Leandro, Albuquerque
Major investments made in past 12 months	Albuquerque facility for radiation effects testing.

generalfusion

FIA MEMBER

GENERAL FUSION

General Fusion's practical Magnetized Target Fusion (MTF) is designed to provide clean and cost-efficient baseload energy to utility and industrial steam heat users. The company operates LM26—a world-first fusion demonstration machine at its Canadian facility. LM26 is targeting transformational near-term technical milestones that position the company to deliver a first-of-a-kind plant producing energy around 2035.

Primary location	Vancouver, Canada
Contact details	info@generalfusion.com
Year founded	2002
Founder names	Dr. Michel Laberge
Primary target market(s)	Electricity generation
Total declared funding	~$500,000,000
General approach	Magnetized Target Fusion (MTF)
Specific approach	Magnetized target fusion
Fuel Source	DT
Planned energy capture approach	Liquid metal with heat exchanger
Target: Net fusion energy gain (scientific)	Lawson criterion by 2028.
Target: Net energy gain (engineering)	First-of-a-Kind plant producing energy by 2035
Target: First pilot plant	Approximately 2035.
Anticipated MWe of your commercial operating facility	Approximately 300MWe from two machines operating in tandem.
Siting agreement for a commercial plant	Engaged end users to join the company's Market Development Advisory Committee (MDAC) on commercialization. Signed MOU with MDAC member Bruce Power in Ontario, Canada, to evaluate potential deployment of a fusion power plant in Ontario.
Milestones in past 12 months	1) Currently executing LM26 demonstration program, compressing large-scale magnetized plasmas with a lithium liner. 2) In June, we announced significant progress toward the company's next major technical milestone with LM26. The results show meaningful plasma heating to electron temperatures of approximately 8.4 million degrees Celsius, or 0.72 keV, driven by the compression of a plasma with a lithium liner. This is a key indicator of success for the company's uniquely practical approach to fusion energy. These results, submitted for peer review and publicly available, demonstrate significant progress toward the key 1 keV electron temperature milestone. 3) General Fusion and Renexia S.p.A., a Toto Group company specializing in renewable energy, announced a framework agreement to advance the commercial deployment of General Fusion's fusion energy technology in Italy. 4) General Fusion was recently ranked first on TIME's prestigious list of the World's Top GreenTech Companies of 2026 for its leadership in fusion energy. 5) At the beginning of 2026, we announced plans to become the first publicly traded pure-play fusion company through a Business Combination Agreement with Spring Valley Acquisition Corp. III
Interim plants or facilities planned prior to your commercial pilot plant?	As we make progress with LM26, we intend to move into our commercialization program, which is an engineering program to design and demonstrate key commercial systems and components such as seals and valves and heat exchange systems.

FIA MEMBER

GENF

GenF's mission is to develop the first inertial confinement nuclear fusion reactor triggered by lasers. A winner of the France 2030 program 2024, GenF is leading the Taranis project in collaboration with CNRS and CEA to harness fusion energy.

Primary location	Elancourt, France
Contact details	contact@genf-systems.com
Year founded	2024
Founder names	Thales Company
Primary target market(s)	Electricity generation
Total declared funding	$10,800,000
Employees	40
General approach	Inertial confinement
Specific approach	Laser-driven inertial confinement
Fuel Source	DT
Planned energy capture approach	Lithium neutron 'blanket'
Target: Net fusion energy gain (scientific)	2035
Target: Net energy gain (engineering)	2037
Target: First pilot plant	2040
Anticipated MWe of your commercial operating facility	1 Gwe
Milestones in past 12 months	1) Validation of high gain (>94) fusion operating point (optimized target design & laser specifications) by CEA codes 2) First cryogenic impregnation of innovative target technology 3) First experimental campaign
Interim plants or facilities planned prior to your commercial pilot plant?	Fusion Hub 2028: demonstrate high frequency inertial fusion pilar by injecting and intercepting with laser a cryogenic target, with frequency > 10 Hz. Progressive intégration of DPSS kJ Laser line
Key collaborators/partners	CEA, CNRS, Polytechnique school, Assystem
Published papers from last 12 months	1) Perspectives in laser-driven inertial fusion reactor system, AIP Advances 2) Implosion and Illumination designs for laser-driven fusion energy, 13th International Conference on Inertial Fusion Sciences and Applications 3) Addressing the first wall challenges for an ICF power plant, 13th International Conference on Inertial Fusion Sciences and Applications

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FIA MEMBER

HELICITYSPACE CORP.

Space Propulsion

Primary location	Pasadena, California, USA
Contact details	info@helicityspace.com
Year founded	2019
Founder names	Dr. Setthivoine You, Marta Calvo, Dr. Stephane Lintner
Primary target market(s)	Space propulsion
Total declared funding	$10,000,000
Employees	9
General approach	Magneto-inertial
Specific approach	Plectoneme
Fuel Source	DD
Planned energy capture approach	Thrust
Target: Net fusion energy gain (scientific)	2032
Target: First pilot plant	2035 Flight in Lower Earth Orbit (LEO)
Anticipated MWe of your commercial operating facility	100kWe first space flight
Siting agreement for a commercial plant	In orbit...
Interim plants or facilities planned prior to your commercial pilot plant?	Test flights with sub scale jet configurations
Key collaborators/partners	Caltech, Los Alamos, NASA, UMBC, Unv. Tokyo.

HELION

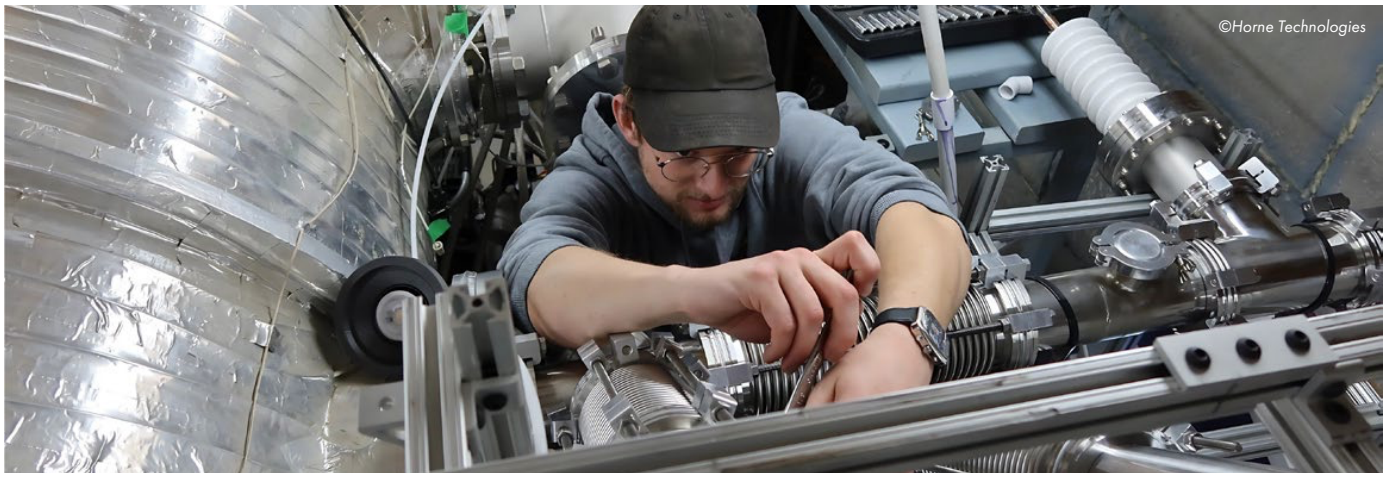
Helion is a fusion energy company focused on generating zero-carbon electricity from fusion. Its mission is to build fusion power plants to fully satisfy the world's need for clean abundant energy.

Primary location	Everett, Washington, USA
Contact details	inquiries@helionenergy.com
Year founded	2013
Founder names	David Kirtley, Chris Pihl, George Votroubek
Primary target market(s)	Electricity generation
Total declared funding	$1,500,000,000
Employees	700
General approach	Magneto-inertial
Specific approach	Field Reversed Configuration
Fuel Source	DHe3
Planned energy capture approach	Direct energy conversion
Target: First pilot plant	2028
Anticipated MWe of your commercial operating facility	Signed a PPA with Microsoft for at least 50 MWe
Siting agreement for a commercial plant	Currently building onsite in Malaga, WA
Power purchase agreement, offtake agreement, or similar commercial offtake commitment	Microsoft, Nucor
Milestones in past 12 months	1) Achieved a first for the private fusion sector by reaching 150 M°C plasma temperatures and demonstrating deuterium-tritium fuel operation in Polaris, Helion's 7th fusion prototype 2) Initiated construction of Orion, the world's first fusion power plant, following receipt of all required environmental approvals and regulatory permits 3) Commissioned a dedicated manufacturing facility for in-house production of pulsed power capacitors, a critical component for Helion's commercial fusion systems
Major investments made in past 12 months	1) Opened Omega, a new 165,000 sq. ft. facility, dedicated to pulsed power manufacturing for Helion's commercial fusion power plants 2) Awarded $4 million across 20 institutions as part of the HERCULES program, advancing technologies for scaled commercial fusion operations 3) Donated more than $70,000 to local organizations in Washington state that support STEM learning and energy education

HORNE TECHNOLOGIES

Horne Technologies is advancing a modular, low-capital path to fusion energy through its dual-confinement hybrid approach. With minimal funding, Horne has validated core technology through multiple reactor prototypes and developed a digital twin to accelerate development. The company is also commercializing a high-flux neutron source for near-term revenue generation.

Primary location	Longmont, Colorado, USA
Contact details	info@hornetech.com
Year founded	2008
Founder names	Tanner Horne
Primary target market(s)	Electricity generation, Off-grid energy, Industrial heat
Total declared funding	$4,772,000
Employees	7
General approach	Hybrid magnetic/electrostatic confinement
Specific approach	Spindle cusp, superconducting shielded-grid IEC
Fuel Source	DD, DT, pB11
Target: First pilot plant	5-10 years
Anticipated MWe of your commercial operating facility	0.1 - 10MWe
Milestones in past 12 months	We have developed a digital twin of our reactor leveraging WarpX and several other advanced modeling and simulation softwares. Horne implemented additional sensors and data collection into it's reactor to supply data for digital twin validation. Stood up partnerships with National Labs.
Interim plants or facilities planned prior to your commercial pilot plant?	Horne technologies will stand up a commercial neutron testing facility for materials development and testing.
Major investments made in past 12 months	Expanded facilities including manufacturing equipment and a clean room. In-house modeling, simulation, and computational infrastructure. Ecosystem development through event coordination
Key collaborators/partners	Lawrence Berkeley National Laboratory, Princeton Plasma Physics Laboratory, University of Wisconsin - Madison, HTS Inc., MetOx Inc.,



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INFROTON FUSION INC

INFROTON has independently demonstrated Deuterium-Deuterium (D-D) fusion using its Magneto-Inertial Fusion (MIF) approach, achieving fusion with only kilojoule-class laser energy. With the underlying physics validated, the company is now focused on engineering a globally scalable, containerized fusion energy platform while developing fusion fuel production and digital energy infrastructure.

Primary location	Darien, Connecticut, USA; EU Research Base: Budapest, Hungary
Contact details	info@infrotonfusion.com
Year founded	2024
Founder names	Dr Andras Pungor, Janos Dobos
Primary target market(s)	Electricity generation, Tritium and helium-3 fusion fuel
Employees	4
General approach	WGM-MIF-CPA
Specific approach	KJ-class whispering gallery mode (WGM) laser-driven cavity pressure acceleration (CPA)-assisted magneto-inertial fusion (MIF)
Fuel Source	DD
Planned energy capture approach	Helium-cooled ceramic pebble bed reactor
Target: Net fusion energy gain (scientific)	March 30, 2027
Target: Net energy gain (engineering)	March 30, 2027
Target: First pilot plant	December 10, 2028
Anticipated MWe of your commercial operating facility	1-35 MWe
Siting agreement for a commercial plant	In discussions with multiple defense companies.
Milestones in past 12 months	1) Validation of WGM-MIF technology: a 100-tesla plasma-compressing magnetic field, and ion and electron temperatures of 169 keV (~1.96 billion °C) in hot spots sustained for 10 ns. 2) Validation of CPA technology: D–D fusion producing 10^9 neutrons and 10^{10} protons at ~3 MeV; hypersonic plasma jet velocity of 0.7×10^8 cm/s. 3) Scaling validation by combining WGM-MIF and CPA technologies: we have secured support from the Laserlab EU Scientific Council for a 7-day experiment in Prague at the PALS laser facility. 4) Validation of the CPA technology: – Demonstration of D–D fusion – ≈10^9 neutrons per shot – ≈10^{10} protons (~3 MeV) – Hypersonic plasma jet velocity: 0.7×10^8 cm/s (≈700 km/s)

KYOTO FUSIONEERING

FIA MEMBER

KYOTO FUSIONEERING

Kyoto Fusioneering (KF) develops the advanced technologies and integrated systems critical for commercial fusion: plasma heating, fusion fuel cycle, and breeding blanket systems. With operations in Japan, the US, the UK and Germany, and a joint venture in Canada, KF partners with fusion developers, national laboratories, and industrial suppliers worldwide to accelerate the realization of fusion energy.

Primary location	(1) Global Headquarters: Tokyo, Japan (2) North American Headquarters: Oak Ridge, Tennessee (3) UK Headquarters: Culham, United Kingdom (4) EU Headquarters: Karlsruhe, Germany
Contact details	info@kyotofusioneering.com
Year founded	2019
Founder names	Dr. Satoshi Konishi, Taka Nagoa, Dr. Richard Pearson, Dr. Shutaro Takeda
Primary target market(s)	Electricity generation, Industrial heat
Total declared funding	$143,854,890
Employees	164
General approach	Agnostic
Specific approach	Agnostic
Fuel Source	Agnostic, with primary investments in DT technology
Planned energy capture approach	Lithium neutron 'blanket'
Milestones in past 12 months	1) Kyoto Fusioneering launched a new R&D Center for Gyrotron Systems in Adogawa, Takashima City, Shiga Prefecture, Japan. 2) Kyoto Fusioneering and Shimadzu Corporation jointly developed a prototype of a turbomolecular pump designed for operation in tritium environments, to be tested and validated on UNITY-2. 3) Kyoto Fusioneering demonstrated reliability of hydrogen recovery for fusion power plants in UNITY-1, using a vacuum sieve tray (VST) technology.
Interim plants or facilities planned prior to your commercial pilot plant?	1) UNITY-1, a blanket component and thermal cycle facility in Kyoto, Japan 2) UNITY-2, a full deuterium-tritium fuel cycle technology demonstration facility located in Chalk River, Canada 3) UNITY-3, which will be housed at ORNL and serve to validate breeding blanket performance with prototypic neutrons. 4) Experimental campaigns with the FLiBe Research Japan Advanced (FREJA) Loop (formerly "Kyoto FLiBe loop") equipped with a FLiBe refining system, and the Kyoto Liquid Lithium Loop (KL3) which is a lab-scale forced convection liquid lithium loop with an inventory of 3.7 kg of molten lithium.
Major investments made in past 12 months	Kyoto Fusioneering has made investments in design and early procurements for UNITY-3, a breeding blanket test facility that will validate tritium breeding blanket performance with prototypic neutrons and component geometries.
Key collaborators/partners	Alloyed Ltd., Canadian Nuclear Laboratory, Fujikura, Idaho National Laboratory, Karlsruhe Institute of Technology, Kyoto University, Kyushu University, Mikuni Jukogyo, Oak Ridge National Laboratory, Pacific Northwest National Laboratory, Princeton Plasma Physics Laboratory, Realta Fusion, Savannah River National Laboratory, Tohoku University, Tsukuba University, TWI Ltd., United Kingdom Atomic Energy Authority, United Kingdom Industrial Fusion Solutions (UKIFS), University of Illinois, University of Tokyo

LASERFUSIONX INC.

LaserFusionX is advancing a unique and very attractive approach to laser fusion energy. We leverage the deep UV light from the argon fluoride (ArF) laser to substantially reduce the laser energy required for high-energy-gain direct drive implosions. This enables the development of smaller lower-cost laser-fusion power plants.

Primary location	Springfield, Virginia, USA
Contact details	LaserFusionX@Outlook.com
Year founded	2022
Founder names	Stephen Obenschain
Primary target market(s)	Electricity generation
Total declared funding	$458,000
Employees	2
General approach	Inertial confinement
Specific approach	Laser-driven inertial confinement
Fuel Source	DT
Planned energy capture approach	Lithium neutron 'blanket'
Target: Net energy gain (engineering)	2040
Target: First pilot plant	2041
Anticipated MWe of your commercial operating facility	425 MWe
Milestones in past 12 months	Publication of high energy ArF laser design
Interim plants or facilities planned prior to your commercial pilot plant?	Construction of full size (30 kJ) ArF beamline. Construction and utilization of 60 kJ ArF facility to refine target physics
Spin outs/patents/commercial innovations	Patent application for reactor chamber design that enables high TBR.
Published papers from last 12 months	"Argon fluoride laser system for high gain direct drive fusion," Opt. Express 33, 48423-48450 (2025)

LONGVIEW FUSION
ENERGY SYSTEMS

FIA MEMBER

LONGVIEW FUSION ENERGY SYSTEMS, INC.

Longview Fusion Energy Systems is building on demonstrated fusion energy gain. While others are still pursuing ignition, Longview will build the world's first laser fusion power plant: a 440 MWe facility operational in the mid-2030s, developed with Fluor and vetted by eight major U.S. utilities. Longview fusion is ready to move from the laboratory to the grid, to behind the meter, and to commodity production.

Primary location	Livermore, California, USA
Contact details	info@longviewfusion.com
Year founded	2021
Founder names	Edward Moses, Aaron Khandros, Igor Khandros
Primary target market(s)	Electricity generation, Medical, Off-grid energy, Hydrogen and/or clean fuels, Industrial heat
Employees	15
General approach	Inertial confinement
Specific approach	Laser-driven inertial confinement
Fuel Source	DT
Planned energy capture approach	Liquid metal with heat exchanger
Anticipated MWe of your commercial operating facility	440MWe-1600MWe, 3000MWt for industrial heat
Milestones in past 12 months	1) Demonstrated fuel capsule acquisition and illumination at plant-relevant parameters. 2) Developed integrated performance, cost and economic models. 3) Advanced the Longview IP portfolio across fusion technologies and plant configuration.
Interim plants or facilities planned prior to your commercial pilot plant?	1) Physics: Based on indirect-drive gain achieved at the National Ignition Facility (Qsci ≈ 4). 2) Rep-rated operation: Integrated laser-target demonstration at full plant repetition rate. 3) Engine operation: Demonstration of line replacement units for cooling and tritium production. 4) Tritium Fuel Processing: Integrated capture, isolation, and storage. 5) Fuel Pellet Manufacturing: Operational production line at full plant demonstration rate.
Key collaborators/partners	DOE: Lawrence Livermore National Laboratory, Savannah River National Laboratory, National Laboratory of the Rockies, and Battelle National Laboratories; EPC: Fluor Corporation; Industrial Partners: Laser and optics companies, target developers, and process designers; Legal and Regulatory: Pillsbury Winthrop Shaw Pittman LLC; Economic analysis: Bates White

MARITIME FUSION

FIA MEMBER

MARITIME FUSION

Low power density tokamaks for marine and off-grid applications

Primary location	San Francisco, California, USA
Contact details	general@maritimefusion.com
Year founded	2025
Founder names	Justin Cohen, Jason Kaufmann
Primary target market(s)	Marine propulsion, Off-grid energy, Hydrogen and/or clean fuels, Industrial heat
Total declared funding	$4,700,000
Employees	6
General approach	Magnetic confinement (Tokamak, Stellarator)
Specific approach	Tokamak
Fuel Source	DT
Planned energy capture approach	Liquid metal with heat exchanger
Target: Net fusion energy gain (scientific)	2032
Target: Net energy gain (engineering)	2032
Target: First pilot plant	2032
Anticipated MWe of your commercial operating facility	25 MWe
Milestones in past 12 months	Demonstrated HTS cable, finalized initial physics basis
Major investments made in past 12 months	REBCO tape
Key collaborators/partners	Columbia University, DIIID Department of Energy
Spin outs/patents/commercial innovations	HTS cable for power distribution, especially for datacenters

MIFTI Fusion

FIA MEMBER

MIFTI FUSION

MIFTI Fusion has pioneered a simple and direct way to generating fusion power using our patented Staged Z-Pinch approach. This involves imploding a high Z-liner on a fusible target via a Z-pinch.

Primary location	Tustin, California, USA
Contact details	contact@miftifusion.com
Year founded	2009
Founder names	Hafiz Rahman, Norman Rostoker, Gerald Simmons, Mohammad Arshad
Primary target market(s)	Electricity generation
Total declared funding	$24,000,000
Employees	8
General approach	Magneto-inertial
Specific approach	Z-pinch
Fuel Source	DT
Planned energy capture approach	Lithium neutron 'blanket'
Target: Net fusion energy gain (scientific)	2028
Target: Net energy gain (engineering)	2030
Target: First pilot plant	2035
Anticipated MWe of your commercial operating facility	50MWe-100MWe
Milestones in past 12 months	1) Experimental demonstration of Z Pinch stability and neutron production on the Double Eagle 3.5MA pulsed power driver. 2) Validation and advanced simulation using the FLASH MHD code. 3) Began design of 12 MA intermediate test facility.
Interim plants or facilities planned prior to your commercial pilot plant?	12 MA pulsed power driver for continued testing at better than breakeven conditions.
Key collaborators/partners	University of California, San Diego (UCSD). Lawrence Livermore National Laboratory (LLNL), FLASH center at the University of Rochester.
Published papers from last 12 months	1) "A two-dimensional numerical study of the magneto-Rayleigh–Taylor instability with FLASH: Application to the staged Z-pinch concept", Phys. Plasmas 32, 033901 (2025) 2) "Investigating radiation effects on liner and target compression dynamics in gas-puff Z-pinches implosions", Phys. Plasmas 32, 123302 (2025) 3) "Staged Z-pinch experiment at the Double-EAGLE pulsed power facility", Plasma Physics and Controlled Fusion

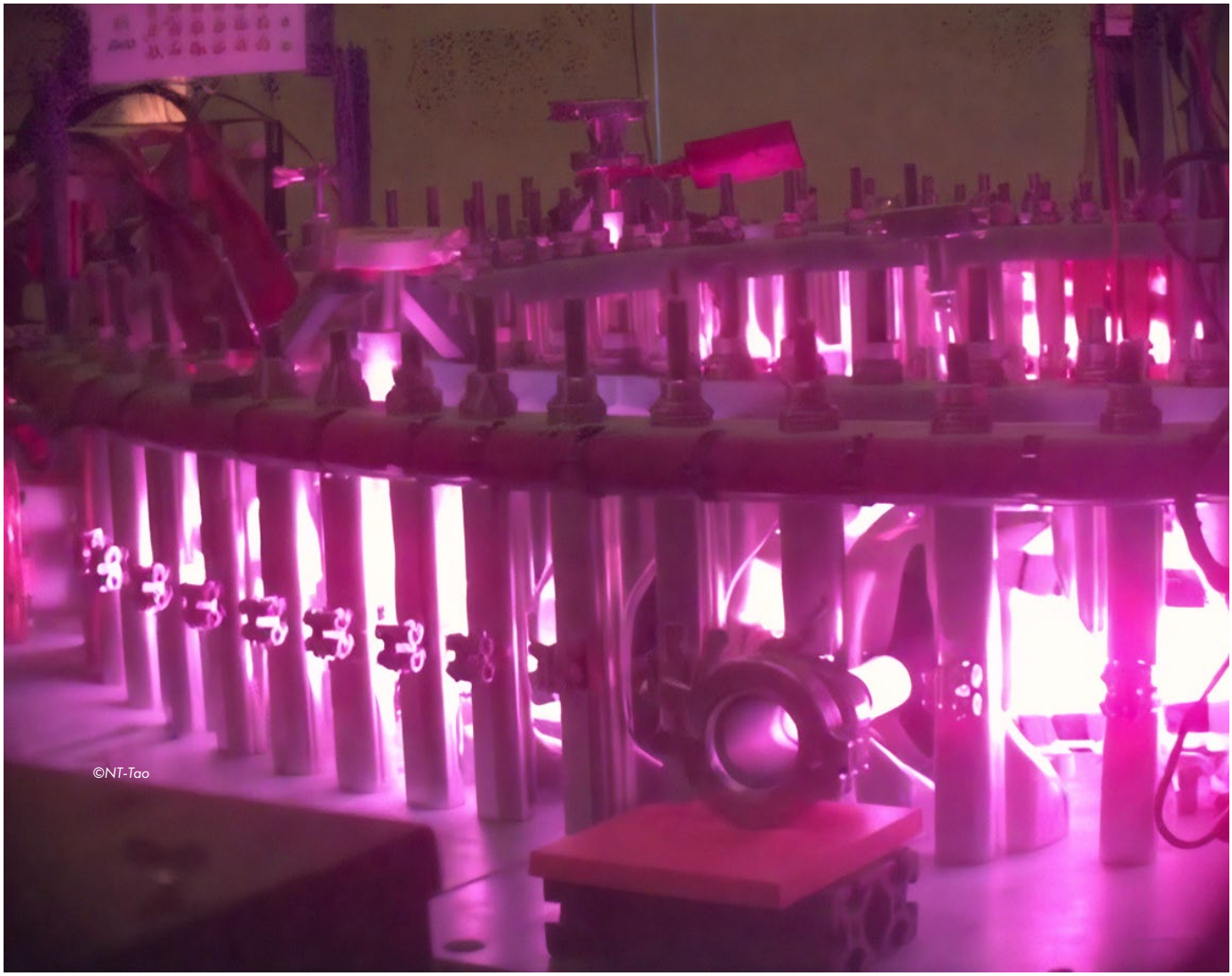
NOVATRON

FIA MEMBER

NOVATRON FUSION GROUP

Fusion power to the grid through development of a mirror machine concept.

Primary location	Stockholm, Sweden
Contact details	info@novatronfusion.com
Year founded	2019
Founder names	Jan Jäderberg & Erik Odén
Primary target market(s)	Electricity generation, Industrial heat
Total declared funding	$34,449,240
Employees	80
General approach	Magnetic Mirror
Specific approach	Open Magnetic confinement
Fuel Source	DT
Planned energy capture approach	Lithium neutron 'blanket'
Target: Net fusion energy gain (scientific)	2030
Target: Net energy gain (engineering)	2035
Target: First pilot plant	2035
Anticipated MWe of your commercial operating facility	1.5 MWe
Milestones in past 12 months	1000+ plasma experiments in Novatron 1, almost doubled number of employees including key strategic recruitments,
Interim plants or facilities planned prior to your commercial pilot plant?	N2 proof of concept, N3 pilot plant
Key collaborators/partners	KTH Royal Institute of Technology, UKAEA, Studsvik, VTT, St1, InnoEnergy, Freemelt, F4E
Spin outs/patents/commercial innovations	Ponderomotive double trap of ions and electrons - a way to confine plasma by ponderomotive force
Published papers from last 12 months	1) Axial confinement in the Novatron Mirror Machine 2) Novatron: Equilibrium and Stability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FIA MEMBER

OPENSTAR TECHNOLOGIES

OpenStar is pursuing the development of levitated dipole fusion reactors for baseload power to the grid. Building upon the groundbreaking experiments of the levitated dipole pioneered by LDX at MIT and RT-1 at the University of Tokyo, OpenStar embraces the natural stability inherent to the dipole field - creating fusion reactors to power the future.

Primary location	Wellington, New Zealand (HQ/facility)
Contact details	info@openstar.tech
Year founded	2021
Founder names	Ratu Mataira-Cole, Thomas Berry
Primary target market(s)	Electricity generation
Total declared funding	$29,400,000
Employees	65
General approach	Magnetic Confinement
Specific approach	Levitated Dipole
Fuel Source	DT
Planned energy capture approach	Lithium neutron 'blanket'
Target: Net fusion energy gain (scientific)	2030
Target: Net energy gain (engineering)	2034
Target: First pilot plant	2034
Anticipated MWe of your commercial operating facility	200 MWe
Milestones in past 12 months	1) Achieved levitated plasma in first machine, Junior, alongside the Prime Minister of NZ 2) Secured $18M USD non-dilutive New Zealand government funding. 3) Published Deuterium-Tritium Levitated Dipole Fusion Power Plants study
Interim plants or facilities planned prior to your commercial pilot plant?	1) Junior [complete] - Demonstrate the elegance and viability of the levitated dipole approach 2) Tahi - Scientific and engineering performance demonstrator 3) Maui - Breakeven device & Volumetric neutron source
Published papers from last 12 months	Deuterium-Tritium Levitated Dipole Fusion Power Plants, in review

PRANOS FUSION ENERGY

FIA MEMBER

PRANOS FUSION PRIVATE LIMITED

Pranos Fusion is an Indian deep-tech company developing low aspect ratio tokamak technology while establishing India's supply chain in fusion-critical verticals of high-temperature superconducting magnets and plasma control systems. These technologies are integrated onto India's first privately developed in-house tokamak, PRAGYA.

Primary location	Bengaluru, India
Contact details	info@pranosfusion.energy
Year founded	2024
Founder names	Dr. Shaurya Kaushal and Roshan George
Primary target market(s)	Electricity generation, Off-grid energy
Total declared funding	$7,217,000
Employees	17
General approach	Magnetic confinement (Tokamak, Stellarator)
Specific approach	Tokamak
Fuel Source	DT
Planned energy capture approach	Both the 'Fission-Fusion Hybrid' & 'Lithium Blanket' concepts remain viable options at this stage.
Target: Net fusion energy gain (scientific)	2030
Target: Net energy gain (engineering)	2034
Anticipated MWe of your commercial operating facility	50 MWe
Milestones in past 12 months	1) PRAGYA: India's first privately built low aspect ratio tokamak, our core prototype for plasma confinement validation - Commenced commissioning and targeting first plasma in 2026. 2) MAGGA (Magnetic Gain): Our in-house magnet development vertical, building HTS magnet technology for compact fusion reactor - Commenced construction of a dedicated magnet development facility encompassing tape testing, characterisation, and coil winding capabilities. 3) JENGA: Our integrated tokamak design, simulation, and digital twin platform built to accelerate fusion device development -Benchmarked core simulation capabilities and initiated collaboration with IPR to integrate JENGA into live tokamak operations, with a focus on plasma control and real-time equilibrium reconstruction.
Major investments made in past 12 months	1) PRAGYA: Gas injection, mechanical handling, microwave heating and power supply systems. 2) MAGGA: Coil winding, high-temperature superconducting (HTS) tape testing and characterisation facility for our magnet technology vertical. 3) JENGA: Second office at the Institute of Plasma Research, Gandhinagar, India to facilitate our collaboration with IPR-tokamaks for integration with JENGA.
Key collaborators/partners	Institute of Plasma Research (IPR), Gandhinagar, Jawaharlal Nehru Centre For Advanced Scientific Research (JNCASR), Bengaluru, Indian Institute of Science (IISc), Bengaluru, Indian Institute of Technology (IIT), Kharagpur.
Published papers from last 12 months	1) Design and mechanical analysis of the PRAGYA tokamak vacuum vessel, [arXiv] 2) Jenga framework for integrated design and modeling of tokamaks: A case study on MAST-U, [arXiv] 3) Electromagnetic design and simulation of the magnet system for the PRAGYA tokamak, [arXiv]

PROXIMA FUSION

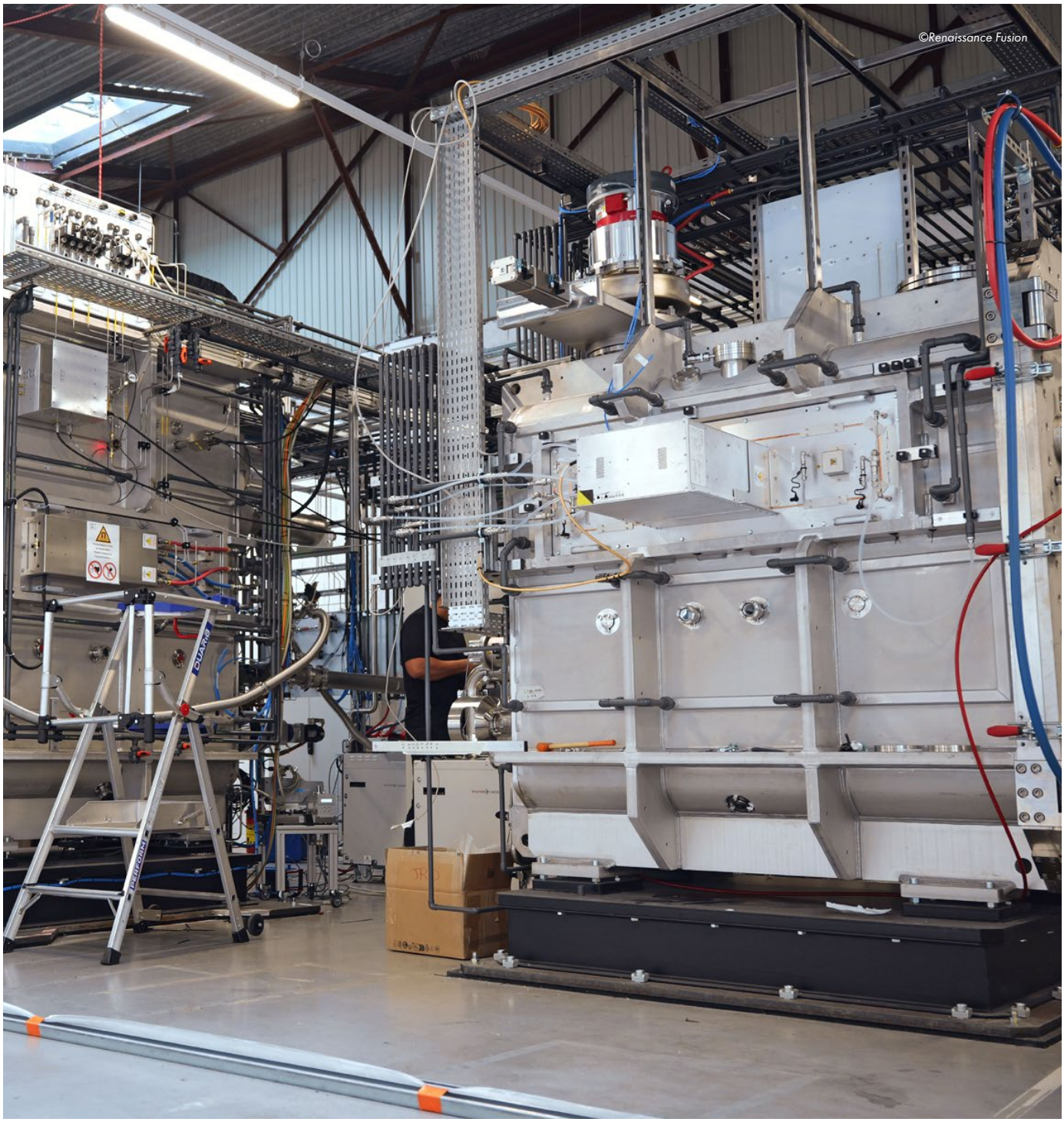
Proxima Fusion is building QI stellarators, leveraging experience from W7-X at the Max Planck Institute for Plasma Physics, together with advances in high-temperature superconducting (HTS) technology and computational optimization.

Primary location	Munich, Germany
Contact details	info@proximafusion.com
Year founded	2023
Founder names	Francesco Sciortino, Lucio Milanese, Jorrit Lion, Jonathan Schilling, Martin Kubie
Primary target market(s)	Electricity generation, Hydrogen and/or clean fuels, Industrial heat
Total declared funding	$752,000,000
Employees	100
General approach	Magnetic confinement (Tokamak, Stellarator)
Specific approach	Stellarator
Fuel Source	DT
Planned energy capture approach	Lithium neutron 'blanket'
Target: Net fusion energy gain (scientific)	2030s
Target: Net energy gain (engineering)	2030s
Target: First pilot plant	2030s
Anticipated MWe of your commercial operating facility	750 MWe
Siting agreement for a commercial plant	Preliminary agreement (MoU) with utility RWE
Milestones in past 12 months	1) Signed MoU with Max Planck Institute for Plasma Physics, Bavarian Government and RWE for the construction of the net-energy-gain stellarator "Alpha" and the FOAK fusion power plant "Stellaris". Bavarian government is committing up to 400M to the realization of Alpha. 2) Manufacturing of HTS non-planar magnet prototypes.
Interim plants or facilities planned prior to your commercial pilot plant?	Construction of "Alpha", a net-energy stellarator in the early 2030s.
Key collaborators/partners	Max Planck Institute for Plasma Physics, Paul Scherrer Institute, Karlsruhe, Institute of Technology, Forschungszentrum Jülich, Bilfinger Nuclear & Energy Transition, University of Wisconsin, Commissariat à l'Énergie, Atomique, University of Bonn, Technical University of Munich, Swiss Plasma Center

REALTA FUSION

Realta Fusion is developing compact, scalable, modular - CoSMo fusion™ - energy systems based on the magnetic mirror concept. Realta offers the lowest capex path to commercial fusion and is targeting industrial heat and power applications including data centers, chemical plants, metal recycling, remote mining, and other heavy industry.

Primary location	Madison, Wisconsin, USA
Contact details	info@realtafusion.com
Year founded	2022
Founder names	Kieran Furlong, Cary Forest, Jay Anderson, Ben Lindley, Oliver Schmitz
Primary target market(s)	Electricity generation, Industrial heat
Total declared funding	$58,800,000
Employees	50
General approach	Magnetic Mirror
Specific approach	Tandem magnetic mirror
Fuel Source	DT
Planned energy capture approach	Lithium neutron 'blanket'
Anticipated MWe of your commercial operating facility	50-500 MWe
Milestones in past 12 months	1) Announced partnerships with Commonwealth Fusion Systems and Kyoto Fusioneering to source HTS magnets and plasma heating devices respectively. 2) Modeled commercially viable energy gain of Q>5 in a tandem magnetic mirror configuration, or Q>10 with a longer center cell. 3) Completed all technical milestones as part of the first budget period of DOE's Milestone-Based Fusion Development Program, and received approval to proceed with second budgetary period. 4) Became the first commercial fusion company to demonstrate direct energy conversion (DEC) of plasma kinetic energy into electricity.
Interim plants or facilities planned prior to your commercial pilot plant?	Realta Fusion will begin operations in its own purpose-built research and development facility, known as The Realta Forge, in 2026.
Key collaborators/partners	US Department of Energy (Office of Science/FES and ARPA-E), University of Wisconsin-Madison, Commonwealth Fusion Systems, Kyoto Fusioneering, Oak Ridge National Laboratory, Lawrence Berkeley National Laboratory, Los Alamos National Laboratory, CompX
Published papers from last 12 months	1) Equilibrium reconstruction in axisymmetric mirrors, Physics of Plasma 2) Confinement performance predictions for a high field axisymmetric tandem mirrors, Journal of Plasma Physics 3) Drift-cyclotron loss-cone instability in 3D simulations of a sloshing-ion simple mirror, Journal of Plasma Physics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SHINE

SHINE is a fusion energy company. Headquartered in Janesville, Wisconsin, we're building toward commercial fusion power through a series of revenue-generating milestones — neutron testing, medical isotopes, and nuclear waste recycling — each one funding the next, and proving the technology that will deliver fusion energy at scale.

Primary location	Janesville, Wisconsin, USA
Contact details	communications@shinefusion.com
Year founded	2010
Founder names	Greg Piefer
Primary target market(s)	Neutron imaging and radiation effects testing for advanced industrial inspection, Medical isotopes, Transmutation of nuclear waste, Electricity generation (long-term)
Total declared funding	$1,100,000,000
Employees	293
General approach	Hybrid electrostatic confinement is closest to what we're planning in future
Specific approach	Test: beam-target, Medical: plasma-target / fission hybrid, Recycling: warm plasma target/fission hybrid, Energy: thermonuclear plasma
Fuel Source	DT
Planned energy capture approach	Fission-fusion hybrid
Target: Net fusion energy gain (scientific)	2035
Target: Net energy gain (engineering)	2040
Target: First pilot plant	Pilot facilities leveraging fusion already commercialized
Milestones in past 12 months	1) Receipt of conditional commitment from EDF for $263M loan to complete the Chyrsalis for domestic Mo-99 production. 2) Design, production, and successful testing of our Plasma Window, an enabling technology upgrade to our core neutron driver system. Designed specifically to fit within and work as an upgrade to our existing neutron driver platform. Validation testing indicates pressures that can increase both neutron strength(fluence) and flux. 3) Completion of EMIS system for Gd-152, Gd-155, Gd-160 enrichment alongside chemical separation for the production of Tb-161 as a therapeutic isotope.
Interim plants or facilities planned prior to your commercial pilot plant?	1) In Wisconsin, Building One houses our steady-state DT fusion source that set a world record in 2019 2) Cassiopeia, produces up to 100,000 patient doses of non-carrier-added lutetium-177 per year 3) Imaging Center provides neutron-based industrial inspection. 4) Our isotope manufacturing and distribution campus is in North Billerica, Massachusetts, making TechneLite Tc-99m generators and related SPECT diagnostic products. 5) Chrysalis, our hybrid fusion-fission medical isotope production facility in Wisconsin is nearing completion and is designed to deliver up to one megawatt of thermal output for commercial molybdenum-99 supply.

STARTORUS FUSION

Startorus Fusion is a Chinese fusion innovator developing compact reactors featuring repetitive pulse operation and magnetic reconnection heating in HTS spherical tokamaks. We adopt a novel fusion technical route to 'Make Tokamaks Simple Again', and thus achieve fusion energy rapidly and cost-effectively.

Primary location	Xi'an, Shaanxi Province, China
Contact details	business@startorus.cn
Year founded	2021
Founder names	Yi TAN, Rui CHEN
Primary target market(s)	Electricity generation, Off-grid energy
Total declared funding	$138,900,000
Employees	154
General approach	Magnetic confinement (Tokamak, Stellarator)
Specific approach	Spherical tokamak
Fuel Source	DT
Planned energy capture approach	Lithium neutron 'blanket'
Target: First pilot plant	2028
Anticipated MWe of your commercial operating facility	100 MWe

TAE TECHNOLOGIES

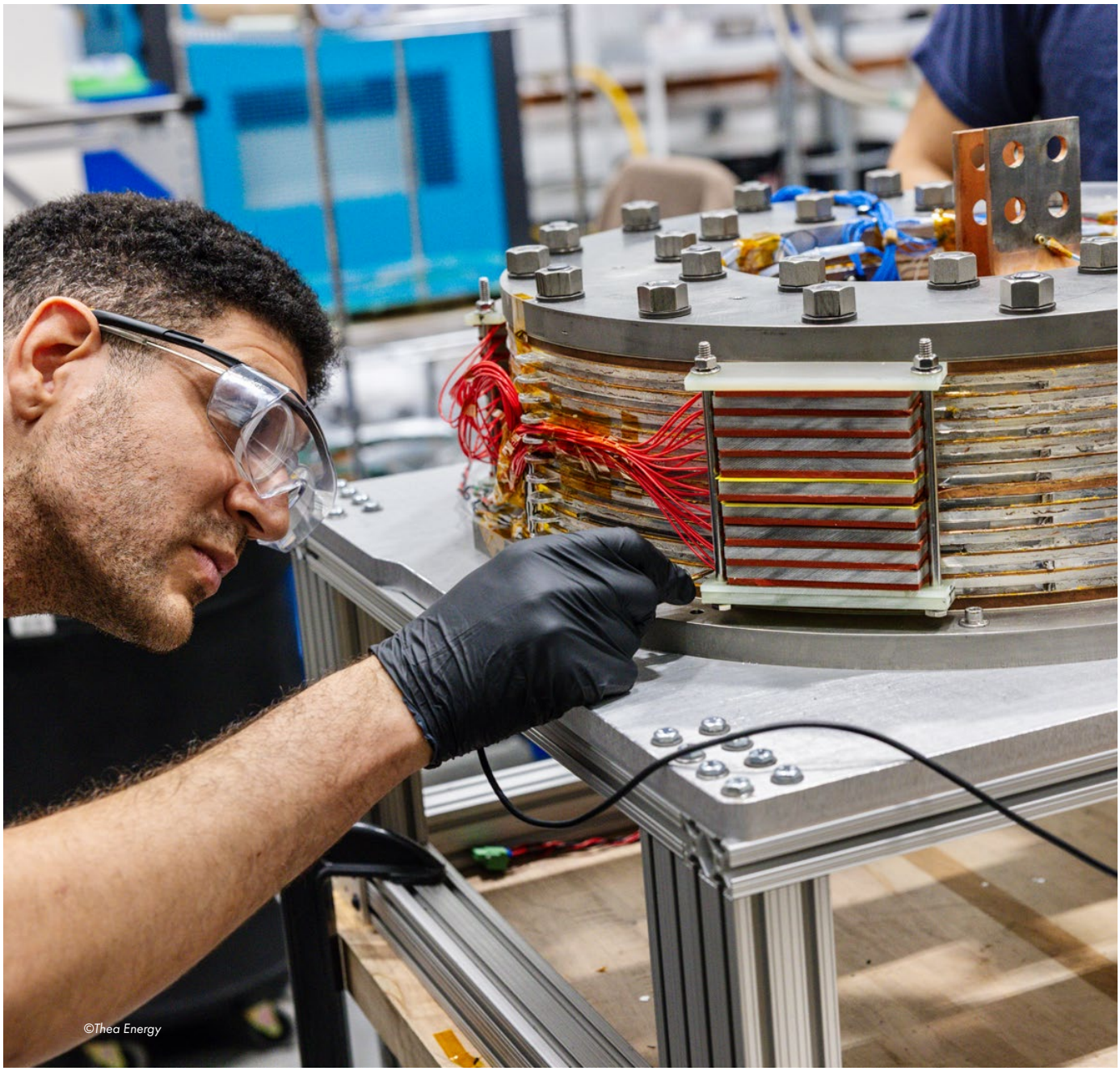
TAE Technologies is developing a sustainable and economically competitive solution to deliver abundant fusion energy to grids and carbon-intensive industries. TAE's proprietary advanced neutral beam-driven FRC platform is designed to enable fuel-agnostic fusion, an approach that has also produced fusion-derived technologies for non-fusion applications. TAE Life Sciences develops technologies and drugs for treating cancer patients. TAE Power Solutions provides power management and energy storage technologies for stationary storage, critical power applications and electric mobility.

Primary location	Foothill Ranch and Irvine, California, USA
Contact details	press@tae.com; pga@tae.com
Year founded	1998
Founder names	Numerous founders
Primary target market(s)	Electricity generation and power management
Total declared funding	$1,500,000,000
Employees	400
General approach	Field-Reversed Configuration
Specific approach	Field-Reversed Configuration
Fuel Source	p-B11
Planned energy capture approach	Heat capture and conventional thermal cycle and/or future direct energy conversion
Target: First pilot plant	2031
Anticipated MWe of your commercial operating facility	50 MWe. Subsequent commercial plants expected to operate in the 350-500 MWe range.
Milestones in past 12 months	Success on TAE's current machine, Norm, has enabled the company to leapfrog its previously planned sixth-generation device, Copernicus, and move directly into the development of its first commercial plant, Da Vinci. Siting process for Da Vinci underway.
Interim plants or facilities planned prior to your commercial pilot plant?	TAE plans to upgrade Norm to explore higher plasma temperature operations while constructing Da Vinci.
Major investments made in past 12 months	1) Neutral beam power upgrade and radio-frequency (RF) power upgrade on Norm to increase plasma temperature and extend confinement time scaling into collisionless regime. 2) TAE Beam UK joint venture with UKAEA to commercialize neutral beam and particle accelerator technology for fusion and non-fusion applications such as food safety, homeland security and targeted radiotherapy for poor-prognosis cancers, currently in development by TAE Life Sciences.
Key collaborators/partners	Argonne National Laboratory, General Atomics, Google, Lawrence Berkeley National Laboratory, Lawrence Livermore National Laboratory, Los Alamos National Laboratory, Massachusetts Institute of Technology, National Institute for Fusion Science – Japan, Nihon University, Oak Ridge National Laboratory, Oxy Low Carbon Ventures, Princeton Plasma Physics Laboratory, Sumitomo Corporation of Americas, Swarthmore College, University of California – Irvine, University of California – Los Angeles, University of Pisa, University of Rochester, University of Texas at Austin, University of Wisconsin – Madison See complete list: https://tae.com/collaborators

TERRA FUSION ENERGY CORPORATION

Terra Fusion Energy Corporation is developing fusion energy systems to provide carbon free, sustainable electrical and thermal energy with engineering simplicity and high reliability. The company builds on the Centrifugal Mirror Fusion Experiment (CMFX), a superconducting magnetic mirror experiment funded by ARPA-E and led by the University of Maryland, Baltimore County (UMBC).

Primary location	Maryland, USA
Contact details	info@tf.energy
Year founded	2024
Founder names	Carlos A. Romero Talamás, Oliver M. Barham
Total declared funding	$1,500,000
Employees	5
General approach	Magnetic confinement
Specific approach	Centrifugal Magnetic Mirror
Fuel Source	DT
Planned energy capture approach	Lead-lithium neutron blanket
Target: First pilot plant	2032
Anticipated MWe of your commercial operating facility	Range of units from 5 MWe to 100 MWe
Major investments made in past 12 months	Established the new company headquarters in Baltimore, Maryland, to develop and test centrifugal mirror components.
Published papers from last 12 months	Measurements of fusion yield on the Centrifugal Mirror Fusion Experiment, 2026 Nucl. Fusion 66, 016010.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TIBBAR PLASMA technologies

FIA MEMBER

TIBBAR PLASMA TECHNOLOGIES, INC.

Our company specializes in Plasma Physics, with particular emphasis on electrostatic confinement fusion machines.

Primary location	Los Alamos, New Mexico, USA
Contact details	r_nebel@hotmail.com
Year founded	2015
Founder names	Richard Nebel
Primary target market(s)	Electricity generation, Space propulsion, Marine propulsion, Off-grid energy
Total declared funding	$4,790,860
Employees	4
General approach	Electrostatic confinement
Specific approach	Magnetic-electrostatic confinement
Fuel Source	pB11
Planned energy capture approach	Direct energy conversion
Target: Net fusion energy gain (scientific)	2035
Target: Net energy gain (engineering)	2035
Target: First pilot plant	2040
Anticipated MWe of your commercial operating facility	50 Mwe
Milestones in past 12 months	1) Producing plasma in the device 2) Demonstrating plasma rotation 3) Observing plasma rings
Interim plants or facilities planned prior to your commercial pilot plant?	Proof of principle device
Published papers from last 12 months	Advanced Fuel Oscillating Fusion Reactor, Fusion Science and Technology

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FIA MEMBER

TYPE ONE ENERGY

Established in 2019 and venture-backed in 2023, Type One Energy is led by a team of globally recognized fusion scientists with a strong track record of building state-of-the-art stellarator fusion machines together with veteran business leaders experienced in scaling companies and commercializing energy technologies.

Primary location	Knoxville, Tennessee, USA
Contact details	thomas.frey@typeoneenergy.com
Year founded	2019
Primary target market(s)	Electricity generation
Total declared funding	$200,000,000
Employees	213
General approach	Magnetic confinement (Tokamak, Stellarator)
Specific approach	Stellarator
Fuel Source	DT
Planned energy capture approach	Lithium neutron 'blanket'
Target: Net energy gain (engineering)	2034
Target: First pilot plant	Break ground on Project Infinity commercial fusion plant in 2026; first power on the grid in 2034.
Anticipated MWe of your commercial operating facility	400 MWe
Siting agreement for a commercial plant	Siting at the Tennessee Valley Authority (TVA) Bull Run Complex, a retired coal power plant in in Clinton, Tennessee
Power purchase agreement, offtake agreement, or similar commercial offtake commitment	September 2025 letter of intent (LOI) with the Tennessee Valley Authority
Milestones in past 12 months	1) In September 2025, The Tennessee Valley Authority (TVA) and Type One Energy announced a Letter of Intent (LOI) to deploy fusion at the Bull Run Energy Complex in East Tennessee. 2) In January, Type One Energy and the Tennessee Valley Authority jointly submitted the initial submission of the "byproduct material license" application for the Bull Run site in preparation for Tennessee's first commercial fusion project. 3) In May, Type One Energy, Tokamak Energy, and AECOM announced the UK Infinity Fusion Consortium to pursue development of the first private-sector-led fusion power plant project in the United Kingdom.
Interim plants or facilities planned prior to your commercial pilot plant?	1) Infinity One is Type One Energy's engineering verification platform at TVA's Bull Run Energy Complex near Oak Ridge, Tenn. 2) FORGED-PFCs is a high-heat flux (HHF) facility at TVA's Bull Run Energy Complex jointly developed with Oak Ridge National Lab and the University of Tennessee, Knoxville (UT).
Major investments made in past 12 months	Type One Energy partnered with the Department of Energy's Oak Ridge National Laboratory (ORNL) and the University of Tennessee, Knoxville (UT) to establish a world-class high-heat flux (HHF) facility, located at the Tennessee Valley Authority's (TVA) Bull Run Energy Complex in East Tennessee, that will evaluate how materials react under extreme conditions in a fusion device.

XCIMER ENERGY CORPORATION

FIA MEMBER

XCIMER ENERGY

We are developing laser-inertial fusion power plants built on low-cost excimer lasers and a liquid-wall chamber that solves material science challenges.

Primary location	Denver, Colorado, USA
Contact details	info@xcimer.energy
Year founded	2021
Founder names	Conner Galloway and Alexander Valys
Primary target market(s)	Electricity generation, Industrial heat
Total declared funding	$178,000,000
Employees	150
General approach	Inertial confinement
Specific approach	Laser-driven inertial confinement
Fuel Source	DT
Planned energy capture approach	Molten salt (Flibe) blanket with heat exchanger
Target: Net fusion energy gain (scientific)	2031
Target: Net energy gain (engineering)	2031
Target: First pilot plant	2035
Anticipated MWe of your commercial operating facility	400 MWe
Milestones in past 12 months	1) Completed and operated kilojoule-scale excimer laser, demonstrating advantages of excimer lasers in manufacturability and scalability. 2) Completed and operated Phoenix demonstration laser facility, the world's largest privately-owned laser system and world's highest-energy utilization of Stimulated Brillouin Scattering. 3) Completed Athena pilot plant pre-conceptual design and submitted to DOE in the Milestone Program.
Interim plants or facilities planned prior to your commercial pilot plant?	1) Anvil will be a 200 kJ laser-fusion beamline demonstrator in 2028. 2) Vulcan will achieve engineering breakeven in 2031 with the world's largest, highest-energy, most powerful and brightest laser (4 MJ, upgradeable to 12 MJ).
Major investments made in past 12 months	Opened a component manufacturing facility in Tucson, AZ. Manufacturing capacitors and other critical components for excimer laser amplifiers.
Key collaborators/partners	Los Alamos National Laboratory, Laboratory for Laser Energetics, US Naval Research Laboratory, Oak Ridge National Laboratory, Savannah River National Laboratory, General Atomics, Westinghouse, MIT, Colorado State University, Texas A&M
Published papers from last 12 months	Stimulated Brillouin scattering spectral profiles and gain coefficients for laser fusion optical drivers, APL Photonics

ZAP ENERGY

Zap Energy is an integrated nuclear energy company building fission, fusion, and intersectional technologies to deliver on-demand, carbon-free power at scale. In fusion, Zap is developing a 50 MW modular core based on sheared-flow-stabilized Z pinch plasmas, which generate fusion energy without external magnets or lasers.

Primary location	Everett, Washington, USA
Contact details	reachout@zap.energy
Year founded	2017
Founder names	Benj Conway, Brian A. Nelson, Uri Shumlak
Primary target market(s)	Electricity generation, Off-grid energy, Industrial heat
Total declared funding	$338,000,000
Employees	150
General approach	Z-Pinch
Specific approach	Sheared-flow-stabilized Z-pinch
Fuel Source	DT
Planned energy capture approach	Liquid metal with heat exchanger
Target: First pilot plant	Late 2030s
Anticipated MWe of your commercial operating facility	200 MWe thermal/50 MWe net electric per module. Multiple modules per facility.
Milestones in past 12 months	1) Produced Zap's first fusion plasmas with pressures exceeding 1 gigapascal in FuZE-3, a new, multi-electrode flow Z-pinch device. 2) Increased performance in Century by 20x to reach 39 kW of average input power delivered in >100 plasma pulses at 0.2 Hz into a circulating liquid metal wall. 3) Completed preconceptual fusion power plant design report describing the full range of systems required to construct and operate a Zap fusion power demonstration facility capable of generating approximately 50 MW of net electrical output.
Interim plants or facilities planned prior to your commercial pilot plant?	1) Zap advances SFS Z-pinch plasma performance toward energy gain on the FuZE series of devices, currently consisting of three FuZE devices in two different facilities, each with dedicated pulsed power supplies and full physics diagnostics suites. 2) Century is Zap's operational 100kW-scale SFS Z-pinch power plant technology demonstrator, validating repetitive pulsed power supplies, plasma-facing circulating liquid metal walls with heat exchange, resilient electrode systems, and power plant technology integration. 3) Millennium is Century's planned successor, designed for 1MW-scale operation. 4) A Demo machine is planned to test operations at commercially-relevant gain conditions prior to the first commercial pilot plant.

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